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Abstract Algebra
Lecture Notes

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Chapter 1

Binary Operations and Groups

1.1 Binary operations

Definition 1.1. A *binary operation* on a set S is a map

$$* : S \times S \longrightarrow S, \qquad (x, y) \longmapsto x * y.$$

It is *associative* if $(x * y) * z = x * (y * z)$ for all $x, y, z \in S$, and *commutative* if $x * y = y * x$ for all $x, y \in S$. An *identity element* is an element $e \in S$ such that $x * e = e * x = x$ for every $x \in S$. Relative to this identity, an *inverse* of x is an element $x' \in S$ such that $x * x' = x' * x = e$.

In additive notation the identity is usually 0 and the inverse of x is $-x$. In multiplicative notation they are 1 or e, and x^{-1}.

Proposition 1.2. *A binary operation has at most one identity element.*

Proof. Suppose e and e' are identities. Then $e = e * e' = e'$.

1.1.1 Congruence classes

Fix a positive integer n. For $a, b \in \mathbb{Z}$, write

$$a \equiv b \pmod{n} \quad \Longleftrightarrow \quad n \mid (a - b).$$

The congruence class of a is

$$[a]_n = a + n\mathbb{Z} = \{a + nk : k \in \mathbb{Z}\}, \qquad \mathbb{Z}_n = \{[a]_n : a \in \mathbb{Z}\}.$$

When n is understood, write $[a]$ for $[a]_n$. Define

$$[a] + [b] = [a + b], \qquad [a][b] = [ab].$$

Exercise 1.1. Show that these two operations on $\mathbb{Z}_n$ are well defined.

Solution. Suppose $a' = a + kn$ and $b' = b + \ell n$. Then

$$(a' + b') - (a + b) = (k + \ell)n,$$

and

$$a'b' - ab = n(a\ell + bk + k\ell n).$$

Thus $[a' + b'] = [a + b]$ and $[a'b'] = [ab]$; neither result depends on the representatives. Associativity and commutativity follow from the corresponding integer identities.

Example 1.3. In $\mathbb{Z}_5$, the five classes are $[0], [1], [2], [3], [4]$, and

$$[1] + [2] = [3], \qquad [3] + [4] = [7] = [2], \qquad [2][4] = [8] = [3].$$

The additive identity is $[0]$, and the multiplicative identity is $[1]$. The additive inverse of $[4]$ is $[1]$; its multiplicative inverse is $[4]$, since $[4]^2 = [16] = [1]$.

Note 1.4. For a finite set with an operation table, commutativity means that the table is symmetric across its main diagonal. For example, addition in $\mathbb{Z}_3$ is given by

$+$	$[0]$	$[1]$	$[2]$
$[0]$	$[0]$	$[1]$	$[2]$
$[1]$	$[1]$	$[2]$	$[0]$
$[2]$	$[2]$	$[0]$	$[1]$

.

1.1.2 Composition of maps

Let $\mathrm{Map}(S, S)$ denote the set of maps from S to itself. Composition gives a binary operation, with convention

$$(f \circ g)(x) = f(g(x)).$$

Exercise 1.2. Prove that composition is associative, and give an example showing that it need not be commutative.

Solution. For every $x \in S$,

$$((f \circ g) \circ h)(x) = f(g(h(x))) = (f \circ (g \circ h))(x).$$

Equality at every input proves equality of the maps. For $S = \{1, 2, 3\}$, take

$$f = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \qquad g = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

Then $(f \circ g)(1) = 1$ while $(g \circ f)(1) = 2$. Hence $f \circ g \neq g \circ f$.

1.2 Groups and basic examples

Definition 1.5. A *group* is a nonempty set G with an associative binary operation, an identity element, and an inverse for every element. A group is *abelian* if its operation is commutative.

We usually write the group operation as multiplication, suppress the operation symbol, and write the identity as e.

Example 1.6. The sets $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are abelian groups under addition, as is $\mathbb{Z}_n$. The sets

$$\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}, \quad \mathbb{R}^\times = \mathbb{R} \setminus \{0\}, \quad \mathbb{C}^\times = \mathbb{C} \setminus \{0\}$$

are abelian groups under multiplication.

Example 1.7. For any set S, its bijections form a group under composition, denoted $\mathrm{Sym}(S)$. Composition is associative, the identity map is the identity element, and the inverse of a bijection is again a bijection. For $S = \{1, \ldots, n\}$ this group is S_n, the *symmetric group*.

Example 1.8. Let $M_n(\mathbb{R})$ be the set of real $n \times n$ matrices. Under addition it is an abelian group. Under multiplication, the invertible matrices form the *general linear group*

$$\mathrm{GL}_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : A \text{ is invertible}\}.$$

Here $(AB)^{-1} = B^{-1}A^{-1}$. For $n = 1$, this group is $\mathbb{R}^\times$.

Exercise 1.3. Show that $\mathrm{GL}_2(\mathbb{R})$ is not abelian.

Solution. Take

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Both determinants are 1, so both matrices are invertible. But

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

which are distinct.

1.3 Elementary group properties

Proposition 1.9. *Every element of a group has a unique inverse.*

Proof. Suppose g' and g'' are inverses of g. Then

$$g' = g'e = g'(gg'') = (g'g)g'' = eg'' = g''.$$

Exercise 1.4. Prove the left and right cancellation laws in a group:

$$ax = ay \Longrightarrow x = y, \qquad xa = ya \Longrightarrow x = y.$$

Solution. Multiplying $ax = ay$ on the left by a^{-1} gives $x = (a^{-1}a)x = (a^{-1}a)y = y$. Multiplying $xa = ya$ on the right by a^{-1} gives $x = y$ in the same way.

Theorem 1.10. *For $a, b \in G$, the equations $ax = b$ and $ya = b$ each have a unique solution:*

$$x = a^{-1}b, \qquad y = ba^{-1}.$$

Proof. Substitution gives $a(a^{-1}b) = b$ and $(ba^{-1})a = b$. Cancellation proves uniqueness.

Note 1.11. These two solutions agree for every a, b if and only if G is abelian. Indeed, equality says $a^{-1}b = ba^{-1}$ for all a, b, and the inverses range over G.

Exercise 1.5. For $a, b \in G$, prove $(ab)^{-1} = b^{-1}a^{-1}$.

Solution. Compute

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = e, \quad (b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b = e.$$

Thus $b^{-1}a^{-1}$ is the unique inverse of ab.

Exercise 1.6. Suppose $a^2 = e$ and $ab = c$ in a group. Show that $a^{-1} = a$ and $ac = b$.

Solution. The equality $a^2 = e$ makes a an inverse of itself, so $a^{-1} = a$. Multiplying $ab = c$ on the left by a gives $ac = a(ab) = (a^2)b = b$.

1.3.1 Nonzero residue classes and units

Example 1.12. Under multiplication, $\mathbb{Z}_5 \setminus \{[0]\}$ is a group:

$$[1]^{-1} = [1], \quad [2]^{-1} = [3], \quad [3]^{-1} = [2], \quad [4]^{-1} = [4].$$

For composite n, the set of nonzero classes need not be closed under multiplication and need not have inverses. For instance, $[2][2] = [0]$ in $\mathbb{Z}_4$, and $[2]x = [1]$ has no solution.

Exercise 1.7. Show that $[a] \in \mathbb{Z}_n$ has a multiplicative inverse if and only if $\gcd(a, n) = 1$. Deduce that the invertible classes form a group $\mathbb{Z}_n^\times$, and that $\mathbb{Z}_p \setminus \{[0]\}$ is a group for prime p.

Solution. If $[a][b] = [1]$, then $ab - kn = 1$ for some $k \in \mathbb{Z}$. Any common divisor of a, n divides 1, so $\gcd(a, n) = 1$. Conversely, Bézout's identity gives $au + nv = 1$ when $\gcd(a, n) = 1$; hence $[u]$ is an inverse of $[a]$. The integer identity used here is proved in the next chapter. The identity $[1]$ is invertible; products of invertible classes are invertible since $(uv)^{-1} = v^{-1}u^{-1}$, and the inverse of an invertible class is invertible. Thus the units form a group. When p is prime, every nonzero residue is relatively prime to p, so these are exactly the nonzero classes.

1.4 Isomorphisms

Definition 1.13. An *isomorphism* between groups $(G, *)$ and $(H, \cdot)$ is a bijection $f : G \to H$ such that

$$f(a * b) = f(a) \cdot f(b) \qquad (a, b \in G).$$

If one exists, write $G \cong H$.

Example 1.14. The map $f : \mathbb{Z}_2 \to \{1, -1\}$ with $f([0]) = 1$ and $f([1]) = -1$ is an isomorphism from addition to multiplication. The operation tables correspond:

$+$	$[0]$	$[1]$
$[0]$	$[0]$	$[1]$
$[1]$	$[1]$	$[0]$

$\cdot$	1	-1
1	1	-1
-1	-1	1

Equivalently, $f([a]) = (-1)^a$; then $f([a] + [b]) = (-1)^{a+b} = f([a])f([b])$.

1.5 The circle and roots of unity

We may represent $\mathbb{Z}_n$ by $\{0, 1, \ldots, n-1\}$, with addition

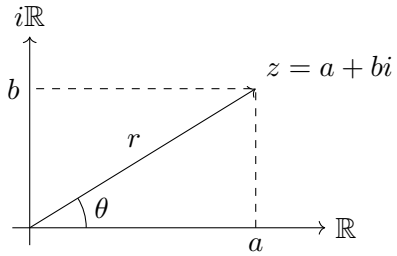
$$a +_n b = \begin{cases} a+b, & a+b < n, \\ a+b-n, & a+b \geq n. \end{cases}$$

For example, $1 +_5 2 = 3$ and $3 +_5 4 = 2$. A continuous counterpart is addition modulo a positive real number c. Define $a \sim b$ if $a - b = kc$ for some $k \in \mathbb{Z}$. The classes form the group $\mathbb{R}/c\mathbb{Z}$ under $[a] + [b] = [a+b]$. Representatives may be chosen in $[0, c)$. For $c = 2\pi$,

$$\pi +_{2\pi} \frac{\pi}{2} = \frac{3\pi}{2}, \qquad \pi +_{2\pi} \pi = 0.$$

For $z = a + bi \neq 0$, write

$$z = re^{i\theta} = r(\cos\theta + i\sin\theta), \qquad r = |z|.$$



Proposition 1.15. *The unit circle $U = \{z \in \mathbb{C} : |z| = 1\}$ is an abelian group under multiplication.*

Proof. The product of two elements has modulus 1. The identity is 1; if $z \in U$, then $z^{-1} = \overline{z}$ also has modulus 1. Associativity and commutativity are inherited from $\mathbb{C}^\times$.

Definition 1.16. For $n \geq 1$, the group of nth roots of unity is

$$U_n = \{z \in \mathbb{C} : z^n = 1\}.$$

If $z^n = 1$, then $|z|^n = 1$, so $|z| = 1$. Write $z = e^{i\theta}$. Then $z^n = 1$ exactly when $n\theta \in 2\pi\mathbb{Z}$. Consequently

$$U_n = \{e^{2\pi ik/n} : k = 0, 1, \ldots, n-1\}.$$

These are n distinct elements; products and inverses remain in U_n.

Example 1.17. For $n = 4$, $U_4 = \{1, -1, i, -i\}$, with multiplication table

$\cdot$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

.

The correspondence $1 \mapsto [0]$, $i \mapsto [1]$, $-1 \mapsto [2]$, $-i \mapsto [3]$ identifies this table with addition in $\mathbb{Z}_4$. In particular, i has order 4.

Exercise 1.8. Prove that $\mathbb{Z}_n \cong U_n$ via $f([a]) = e^{2\pi ia/n}$.

Solution. If $a - b = kn$, then $e^{2\pi ia/n}/e^{2\pi ib/n} = e^{2\pi ik} = 1$, so the map is well defined. Every root has the displayed form, so f is onto. If $f([a]) = f([b])$, then $(a-b)/n \in \mathbb{Z}$, giving $[a] = [b]$. Finally,

$$f([a] + [b]) = e^{2\pi i(a+b)/n} = e^{2\pi ia/n} e^{2\pi ib/n} = f([a])f([b]).$$

Thus f is a bijection preserving the operation.

Exercise 1.9. For $c > 0$, prove that $\mathbb{R}/c\mathbb{Z} \cong U$ under $[a] \mapsto e^{2\pi ia/c}$.

Solution. The exponential values agree exactly when $(a-b)/c \in \mathbb{Z}$, which is exactly the equality of classes in $\mathbb{R}/c\mathbb{Z}$. This gives both well-definedness and injectivity. Each $e^{i\theta} \in U$ is the image of $[c\theta/(2\pi)]$, so the map is onto. The exponential identity $e^{s+t} = e^s e^t$ shows that addition of classes corresponds to multiplication.

Chapter 2

Subgroups and Cyclic Groups

2.1 Subgroup tests

Definition 2.1. A *subgroup* H of a group G is a subset that is itself a group under the operation inherited from G. Write $H \leq G$.

Exercise 2.1. Show that a nonempty subset $H \subseteq G$ is a subgroup if and only if $xy^{-1} \in H$ for all $x, y \in H$.

Solution. The condition holds in every subgroup. Conversely, choose $h \in H$. Then $e = hh^{-1} \in H$, and $y^{-1} = ey^{-1} \in H$ for each $y \in H$. Now $xy = x(y^{-1})^{-1} \in H$. Associativity is inherited from G, so all group axioms hold in H.

Exercise 2.2. Let H be a finite nonempty subset of a group. Prove that H is a subgroup if and only if it is closed under multiplication.

Solution. Only the reverse direction needs proof. For $h \in H$, all positive powers of h lie in H. Finiteness gives $h^i = h^j$ for some $1 \leq i < j$. Cancellation in G yields $h^{j-i} = e \in H$. If $j - i \geq 2$, then $h^{j-i-1} \in H$ is an inverse of h; if $j - i = 1$, then $h = e$, whose inverse is already in H. Thus H contains the identity and all inverses, so it is a subgroup.

Note 2.2. Finiteness is essential: the positive integers are closed under addition in $\mathbb{Z}$, but they do not form a subgroup.

2.2 Cyclic subgroups

For $a \in G$, define $a^0 = e$, the usual positive powers, and $a^{-n} = (a^{-1})^n$ for $n > 0$. Then

$$a^m a^n = a^{m+n}, \qquad (a^m)^{-1} = a^{-m}, \qquad (a^m)^n = a^{mn} \quad (m, n \in \mathbb{Z}).$$

Definition 2.3. The *cyclic subgroup generated by* a is

$$\langle a \rangle = \{a^n : n \in \mathbb{Z}\}.$$

The group G is *cyclic* if $G = \langle a \rangle$ for some $a \in G$, called a *generator*.

The power laws show directly that $\langle a\rangle \le G$. Any subgroup containing a contains all its positive and negative powers. Hence $\langle a\rangle$ is the smallest subgroup containing a.

Example 2.4. In additive notation, $\mathbb{Z} = \langle 1\rangle = \langle -1\rangle$ and $\langle 2\rangle = 2\mathbb{Z}$. In $\mathbb{Z}_4$, the generators are $[1]$ and $[3]$, whereas $\langle [2]\rangle = \{[0],[2]\}$. Correspondingly, U_4 is generated by either i or $-i$.

Lemma 2.5. *Every cyclic group is abelian.*

Proof. If $G = \langle a\rangle$, write $g = a^m$ and $h = a^n$. Then $gh = a^{m+n} = a^{n+m} = hg$.

Theorem 2.6. *Every subgroup of a cyclic group is cyclic.*

Proof. Suppose $G = \langle a\rangle$ and $H \le G$. If $H = \{e\}$, then $H = \langle e\rangle$. Otherwise, H contains a nonidentity power a^k; replacing k by $-k$ if necessary shows that $\{m > 0 : a^m \in H\}$ is nonempty. Let m be its least element. Since $a^m \in H$, we have $\langle a^m\rangle \subseteq H$. For $a^n \in H$, divide

$$n = qm + r, \qquad 0 \le r < m.$$

Then $a^r = a^n(a^m)^{-q} \in H$. Minimality of m forces $r = 0$, so $a^n = (a^m)^q$. Thus $H = \langle a^m\rangle$.

The integer division used above allows negative n as well: for example,

$$38 = 7 \cdot 5 + 3, \qquad -22 = 7 \cdot (-4) + 6.$$

Corollary 2.7. *Every subgroup of $(\mathbb{Z}, +)$ is $d\mathbb{Z}$ for a unique integer $d \ge 0$.*

Proof. Apply the theorem to $\mathbb{Z} = \langle 1\rangle$. The trivial subgroup is $0\mathbb{Z}$; otherwise d is its least positive element, which also gives uniqueness.

2.3 Greatest common divisors and Bézout's identity

Theorem 2.8. *For integers r, s, not both zero, let $d = \gcd(r, s) > 0$. Then*

$$\{nr + ms : n, m \in \mathbb{Z}\} = d\mathbb{Z}.$$

In particular, there exist $u, v \in \mathbb{Z}$ such that $ur + vs = d$.

Proof. The set $H = \{nr + ms : n, m \in \mathbb{Z}\}$ is a nonzero additive subgroup of $\mathbb{Z}$: it contains 0, and differences of its elements have the same form. Thus $H = d_0\mathbb{Z}$ for some $d_0 > 0$. Since $r, s \in H$, d_0 divides both r and s, so $d_0 \mid d$. Conversely, $d_0 \in H$ gives $d_0 = ur + vs$, and d divides this combination. Hence $d \mid d_0$, so $d = d_0$.

Note 2.9. Writing this subgroup as $\langle r, s\rangle$, we have $\gcd(r, s) = 1$ if and only if $\langle r, s\rangle = \mathbb{Z}$.

Example 2.10. The Euclidean algorithm gives

$$72 = 42 + 30, \quad 42 = 30 + 12, \quad 30 = 2 \cdot 12 + 6, \quad 12 = 2 \cdot 6.$$

Thus $\gcd(42, 72) = 6$, and back substitution gives

$$6 = 30 - 2(42 - 30) = 3 \cdot 72 - 5 \cdot 42.$$

2.4 Orders and classification of cyclic groups

Definition 2.11. The *order* of a finite group G is its number of elements $|G|$. An infinite group has infinite order. The *order of an element* $a \in G$, denoted $\operatorname{ord}(a)$, is $|\langle a\rangle|$.

For example, $|\mathbb{Z}_n| = |U_n| = n$, while $\mathbb{Z}$, U, and $d\mathbb{Z}$ for $d > 0$ are infinite.

Theorem 2.12. *Let $G = \langle a\rangle$. If G is infinite, then $G \cong (\mathbb{Z}, +)$. If G is finite, let n be the smallest positive integer with $a^n = e$. Then $|G| = n$ and $G \cong (\mathbb{Z}_n, +)$.*

Proof. If no positive power of a is e, equality $a^k = a^\ell$ implies $a^{k-\ell} = e$, hence $k = \ell$. The map $a^k \mapsto k$ is therefore well defined and bijective; moreover,

$$a^k a^\ell \longmapsto k + \ell.$$

It is an isomorphism onto $\mathbb{Z}$.

If G is finite, two powers coincide, so a positive power is e. Let n be least. For any $k \in \mathbb{Z}$, division $k = qn + r$, $0 \le r < n$, gives $a^k = a^r$. The elements $e, a, \ldots, a^{n-1}$ are distinct: otherwise a positive power smaller than n would be e. Thus $|G| = n$. More generally,

$$a^k = a^\ell \quad \Longleftrightarrow \quad n \mid (k - \ell),$$

as division of $k - \ell$ by n shows. Hence $a^k \mapsto [k]_n$ is a well-defined bijection. Preservation of the operation is verified in the following exercise.

Exercise 2.3. Complete the finite case by proving that $\psi : G \to \mathbb{Z}_n$, $\psi(a^k) = [k]_n$, preserves the group operation.

Solution. For any integers k, ℓ,

$$\psi(a^k a^\ell) = \psi(a^{k+\ell}) = [k + \ell]_n = [k]_n + [\ell]_n = \psi(a^k) + \psi(a^\ell).$$

Together with the established bijectivity, this proves $G \cong \mathbb{Z}_n$.

Corollary 2.13. *If $\operatorname{ord}(a) = n < \infty$, then $a^k = e$ if and only if $n \mid k$. In particular,*

$$\operatorname{ord}(a) = \min\{k > 0 : a^k = e\}.$$

Proof. The power criterion is the equality test in the finite part of the theorem, with $\ell = 0$.

Example 2.14. If $G = \langle a\rangle$ is infinite cyclic, its subgroups are exactly $\langle a^k\rangle$ for $k \ge 0$. Under $a^m \mapsto m$, these correspond to $k\mathbb{Z}$; $k = 0$ gives the trivial subgroup.

2.5 Subgroups of finite cyclic groups

Theorem 2.15. *Suppose $G = \langle a\rangle$ has order n. For $s \in \mathbb{Z}$, put $d = \gcd(n, s)$. Then*

$$\operatorname{ord}(a^s) = \frac{n}{d}, \qquad \langle a^s\rangle = \langle a^d\rangle.$$

Consequently,

$$\langle a^s\rangle = \langle a^t\rangle \quad \Longleftrightarrow \quad \gcd(n, s) = \gcd(n, t).$$

Proof. Write $n = dn'$ and $s = ds'$ with $\gcd(n', s') = 1$. Then

$$(a^s)^m = e \quad \Longleftrightarrow \quad n \mid sm \quad \Longleftrightarrow \quad n' \mid s'm \quad \Longleftrightarrow \quad n' \mid m.$$

For the last implication, choose u, v with $un' + vs' = 1$ and multiply by m. Thus the least positive such m is $n' = n/d$. Bézout's identity also gives $d = un + vs$, so

$$a^d = (a^n)^u (a^s)^v = (a^s)^v \in \langle a^s \rangle.$$

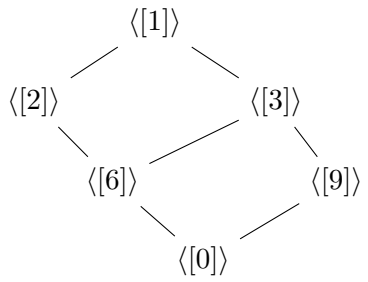
Conversely, $a^s = (a^d)^{s/d}$ lies in $\langle a^d \rangle$. This proves the subgroup identity. If the two gcds agree, their subgroups agree; if the subgroups agree, their orders $n/\gcd(n, s)$ and $n/\gcd(n, t)$ agree, forcing equality of the gcds.

Example 2.16. In $(\mathbb{Z}_{24}, +)$, $\operatorname{ord}([4]) = 24/\gcd(24, 4) = 6$. More generally, $[s]$ generates $\mathbb{Z}_n$ if and only if $\gcd(n, s) = 1$.

Example 2.17. All subgroups of $\mathbb{Z}_{18}$ are

$$\langle [1] \rangle, \quad \langle [2] \rangle, \quad \langle [3] \rangle, \quad \langle [6] \rangle, \quad \langle [9] \rangle, \quad \langle [0] \rangle.$$

Their orders are $18, 9, 6, 3, 2, 1$, respectively. Indeed, each generated subgroup is $\langle [d] \rangle$ for a divisor d of 18, and $\langle [18] \rangle = \langle [0] \rangle$.



A line in this diagram joins a subgroup to an immediate larger subgroup above it.

Exercise 2.4. Let d_1, d_2 be positive divisors of n, and let $G = \langle a \rangle$ have order n. Show that

$$\langle a^{d_1} \rangle \subseteq \langle a^{d_2} \rangle \quad \Longleftrightarrow \quad d_2 \mid d_1.$$

Solution. If $d_2 \mid d_1$, then a^{d_1} is a power of a^{d_2}, so inclusion follows. Conversely, inclusion gives $a^{d_1} = a^{kd_2}$ for some $k \in \mathbb{Z}$, hence $n \mid (d_1 - kd_2)$. Since $d_2 \mid n$, it follows that $d_2 \mid d_1$.

Chapter 3

Homomorphisms and Cyclic Images

3.1 Homomorphisms, images, and inverse images

Definition 3.1. A *group homomorphism* from a group G to a group H is a map $f : G \to H$ satisfying $f(xy) = f(x)f(y)$ for every $x, y \in G$. For $A \subseteq G$ and $B \subseteq H$, write

$$f(A) = \{f(a) : a \in A\}, \qquad f^{-1}(B) = \{x \in G : f(x) \in B\}.$$

In particular, $f^{-1}(h)$ means the set $\{x \in G : f(x) = h\}$; it does not assert that f has an inverse function. The *image* and *kernel* are

$$\operatorname{im} f = f(G), \qquad \ker f = f^{-1}(e_H).$$

Exercise 3.1. Establish the basic properties of a homomorphism $f : G \to H$: it preserves the identity, inverses, and integer powers; images of subgroups and inverse images of subgroups are subgroups. Show also that f is injective if and only if $\ker f = \{e_G\}$.

Solution. The identity $f(e_G) = f(e_G)^2$ gives $f(e_G) = e_H$ by cancellation. Since $f(x)f(x^{-1}) = f(e_G) = e_H$, we have $f(x^{-1}) = f(x)^{-1}$. Induction proves $f(x^k) = f(x)^k$ for $k \geq 0$, and the inverse identity proves it for negative k. If $K \leq G$, then $e_H \in f(K)$ and $f(x)f(y)^{-1} = f(xy^{-1}) \in f(K)$ for $x, y \in K$; hence $f(K) \leq H$. If $L \leq H$, then $e_G \in f^{-1}(L)$, and for $x, y \in f^{-1}(L)$, $f(xy^{-1}) = f(x)f(y)^{-1} \in L$. Thus $f^{-1}(L) \leq G$. In particular, $\operatorname{im} f \leq H$ and $\ker f \leq G$. An injective map has only e_G mapping to e_H. Conversely, if the kernel is trivial and $f(x) = f(y)$, then $f(xy^{-1}) = e_H$, so $xy^{-1} = e_G$ and $x = y$.

Exercise 3.2. If $f : G \to H$ and $g : H \to K$ are homomorphisms, prove that $g \circ f : G \to K$ is a homomorphism.

Solution. For $x, y \in G$,

$$(g \circ f)(xy) = g(f(x)f(y)) = g(f(x))g(f(y)) = (g \circ f)(x)(g \circ f)(y).$$

3.2 Maps determined by a generator

For a cyclic group $\langle a\rangle$ of finite order n, the formulas

$$\langle a^s\rangle = \langle a^{\gcd(n,s)}\rangle, \qquad \operatorname{ord}(a^s) = \frac{n}{\gcd(n,s)}$$

show that a^s generates the whole group exactly when $\gcd(n,s)=1$, and that $a^s = e$ exactly when $n \mid s$. In additive notation, $\operatorname{ord}([s]_n) = n/\gcd(n,s)$.

Lemma 3.2. *Let $G = \langle a\rangle$ have order n, and let $h \in H$. A homomorphism $f : G \to H$ with $f(a) = h$ exists if and only if $h^n = e_H$. When it exists, it is uniquely determined by*

$$f(a^s) = h^s \qquad (s \in \mathbb{Z}).$$

If h has finite order m, this condition is equivalent to $m \mid n$. In particular, if $H = \langle b\rangle$ has order m, the assignment $a \mapsto b^k$ defines a homomorphism exactly when $m \mid nk$.

Proof. A homomorphism must satisfy $f(a^s) = f(a)^s$ and $e_H = f(a^n) = h^n$. Conversely, suppose $h^n = e_H$. If $a^s = a^t$, then $n \mid s-t$, so $h^s = h^t$; thus the stated formula is well defined. Moreover,

$$f(a^s a^t) = h^{s+t} = h^s h^t = f(a^s) f(a^t).$$

For an element of order m, division of n by m shows that $h^n = e_H$ exactly when $m \mid n$. Taking $h = b^k$ gives the final assertion.

Example 3.3. For the groups of complex roots of unity, the map

$$U_{28} \longrightarrow U_7, \qquad w \longmapsto w^4$$

is a homomorphism: $(w^4)^7 = w^{28} = 1$ and $(uv)^4 = u^4 v^4$. If z generates U_{28}, then z^4 has order 7, so this map is surjective. Its kernel is $U_4 \subset U_{28}$. Another example is

$$\mathbb{Z}_{21} \longrightarrow \mathbb{Z}_7, \qquad [a]_{21} \longmapsto [4a]_7.$$

Changing a by $21t$ changes $4a$ by $84t$, a multiple of 7, so the map is well defined; preservation of addition follows from $4(a+b) = 4a + 4b$.

Example 3.4. The proposed assignment $[1]_4 \mapsto [2]_3$ cannot define a homomorphism $\mathbb{Z}_4 \to \mathbb{Z}_3$. Such a homomorphism would give

$$[0]_3 = f(4[1]_4) = 4f([1]_4) = 4[2]_3 = [2]_3,$$

a contradiction.

Exercise 3.3. Describe every homomorphism $\mathbb{Z}_8 \to \mathbb{Z}_{12}$.

Solution. The image of $[1]_8$ must be $[k]_{12}$ with $12 \mid 8k$, equivalently $3 \mid k$. Thus there are exactly four maps:

$$f_k([a]_8) = [ka]_{12}, \qquad k \in \{0,3,6,9\}.$$

Their images are respectively $\{0\}$, $\{0,3,6,9\}$, $\{0,6\}$, and $\{0,3,6,9\}$ in $\mathbb{Z}_{12}$.

Chapter 4

Permutation Groups and Parity

4.1 Notation, composition, and cycles

Definition 4.1. A *permutation* of a set X is a bijection $X \to X$. The permutations form the *symmetric group* $\mathrm{Sym}(X)$ under composition. For $X = \{1, \dots, n\}$ write S_n. A *permutation group* on X is a subgroup of $\mathrm{Sym}(X)$. Products act from right to left: $(\sigma\tau)(i) = \sigma(\tau(i))$.

Example 4.2. Consider

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 & 1 \end{pmatrix}, \qquad \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix}.$$

Then

$$(\sigma\tau)(1) = \sigma(3) = 3, \qquad (\sigma\tau)(3) = \sigma(1) = 5,$$

and the complete product is

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 1 & 2 \end{pmatrix} = (1\ 3\ 5\ 2\ 4).$$

Definition 4.3. For distinct elements $a_1, \dots, a_r$, the *cycle* $(a_1\ a_2\ \cdots\ a_r)$ sends a_i to a_{i+1}, sends a_r to a_1, and fixes every other element. Its length is r. Two cycles are *disjoint* if their sets of displayed elements are disjoint. A cycle of length two is a *transposition*.

Example 4.4. In S_6,

$$(1\ 3\ 5) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 5 & 4 & 1 & 6 \end{pmatrix}.$$

In S_{10},

$$(2\ 4\ 6\ 7) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 1 & 4 & 3 & 6 & 5 & 7 & 2 & 8 & 9 & 10 \end{pmatrix}.$$

The latter cycle moves four elements and fixes $1, 3, 5, 8, 9, 10$.

Definition 4.5. For $\sigma \in S_n$, define $i \sim j$ if $\sigma^k(i) = j$ for some $k \in \mathbb{Z}$. The equivalence class of i is its *orbit* under σ:

$$\mathrm{Orb}_\sigma(i) = \{\sigma^k(i) : k \in \mathbb{Z}\}.$$

Exercise 4.1. Prove that this relation is an equivalence relation, and that every permutation is a product of disjoint cycles, uniquely up to the order of the cycles and cyclic rotation within each cycle. Fixed points may be omitted.

Solution. Reflexivity uses $k = 0$. If $j = \sigma^k(i)$, then $i = \sigma^{-k}(j)$, proving symmetry. If also $\ell = \sigma^t(j)$, then $\ell = \sigma^{t+k}(i)$, proving transitivity. Starting from i, finiteness forces a repeated value in $i, \sigma(i), \sigma^2(i), \ldots$. Applying an inverse power to a repetition shows that the first return is to i. If r is the least positive return time, the orbit gives the cycle

$$(i\ \sigma(i)\ \cdots\ \sigma^{r-1}(i)).$$

Different orbits are disjoint. The product of these cycles agrees with σ on every element. Disjoint cycles commute, because a cycle fixes the support of every disjoint cycle. Finally, the orbits are determined by σ, and its action determines the successor of every entry in each orbit. This proves the stated uniqueness.

Example 4.6. For $(1\ 2)(3\ 4) \in S_4$, the orbits are $\{1, 2\}$ and $\{3, 4\}$. For $(1\ 2\ 3) \in S_5$, they are $\{1, 2, 3\}$, $\{4\}$, and $\{5\}$.

4.2 The regular permutation representation

Definition 4.7. For $a \in G$, define left and right multiplication maps

$$\lambda_a : G \to G, \quad x \mapsto ax, \qquad \rho_a : G \to G, \quad x \mapsto xa.$$

They are bijections, with inverses $\lambda_{a^{-1}}$ and $\rho_{a^{-1}}$.

Theorem 4.8 (Cayley's theorem)**.** *Every group G is isomorphic to a subgroup of* $\mathrm{Sym}(G)$*. If G is finite of order n, it is therefore isomorphic to a subgroup of S_n.*

Proof. Define $\Phi : G \to \mathrm{Sym}(G)$ by $\Phi(a) = \lambda_a$. For $a, b, x \in G$,

$$\lambda_{ab}(x) = (ab)x = a(bx) = (\lambda_a\lambda_b)(x),$$

so Φ is a homomorphism. If $\lambda_a = \lambda_b$, evaluation at e_G gives $a = b$; hence Φ is injective and identifies G with its image. Labeling the elements of a finite G by $1, \ldots, n$ identifies $\mathrm{Sym}(G)$ with S_n.

Note 4.9. Right multiplication satisfies $\rho_a\rho_b = \rho_{ba}$. Thus left multiplication supplies the homomorphism in Cayley's theorem with the composition convention used here.

Exercise 4.2. Write Cayley's embedding explicitly for the Klein four group $V = \{e, a, b, c\}$, where $a^2 = b^2 = c^2 = e$ and $ab = ba = c$.

Solution. Label e, a, b, c by $1, 2, 3, 4$. Left multiplication gives

$$e \longmapsto \mathrm{id}, \qquad a \longmapsto (1\ 2)(3\ 4), \qquad b \longmapsto (1\ 3)(2\ 4), \qquad c \longmapsto (1\ 4)(2\ 3).$$

For example, c exchanges e with c and a with b. These four permutations are distinct, and multiplication agrees with that of V by the identity $\lambda_{xy} = \lambda_x\lambda_y$.

4.3 Transpositions and well-defined parity

Theorem 4.10. *Every element of S_n is a product of transpositions.*

Proof. Using the disjoint-cycle decomposition, it suffices to treat one cycle. Direct evaluation gives

$$(a_1\ a_2\ \cdots\ a_r) = (a_1\ a_2)(a_2\ a_3)\cdots(a_{r-1}\ a_r).$$

Thus an r-cycle is a product of $r-1$ transpositions. The identity is the empty product.

Lemma 4.11. *The identity permutation cannot be a product of an odd number of transpositions.*

Proof. Suppose otherwise and choose the least odd length L of such a product. Fix a letter a that occurs in one length-L expression for the identity. Among all length-L expressions for the identity in which a occurs, choose one with the fewest occurrences of a. The letter a cannot occur only once: if the expression is $P(a\ b)Q$ with P, Q fixing a, it sends a to $P(b) \neq a$.

Cyclically rotating the factors of an identity product still gives an identity product, so arrange that the first factor is $(a\ b)$. Let $(a\ x)$ be the next factor containing a. Move it left across all intervening transpositions. If τ fixes a, then

$$\tau(a\ x) = (a\ \tau(x))\tau.$$

Consequently it can be moved into the second position, with no change in the number of occurrences of a. The product now begins $(a\ b)(a\ c)$. If $b = c$, these factors cancel, giving an identity product of the smaller odd length $L-2$, a contradiction. If $b \neq c$, use

$$(a\ b)(a\ c) = (a\ c)(c\ b).$$

This leaves the length unchanged but reduces the number of occurrences of a by one, while retaining at least one occurrence. It contradicts the minimal choice. Both cases are impossible.

Theorem 4.12. *If a permutation is expressed as a product of r transpositions and also as a product of s transpositions, then r and s have the same parity.*

Proof. Write $\sigma = t_1 \cdots t_r = u_1 \cdots u_s$. Since each transposition is its own inverse,

$$\mathrm{id} = t_1 \cdots t_r u_s \cdots u_1.$$

The lemma implies that $r + s$ is even, which is equivalent to r and s having the same parity.

Definition 4.13. A permutation is *even* if it is a product of an even number of transpositions, and *odd* if it is a product of an odd number. Its *sign* is

$$\operatorname{sgn}(\sigma) = \begin{cases} 1, & \sigma \text{ is even}, \\ -1, & \sigma \text{ is odd}. \end{cases}$$

Exercise 4.3. Prove that $\operatorname{sgn} : S_n \to \{1, -1\}$ is a homomorphism, where the target uses multiplication.

Solution. Concatenating expressions of lengths r and s for σ and τ gives an expression of length $r + s$ for $\sigma\tau$. Therefore

$$\operatorname{sgn}(\sigma\tau) = (-1)^{r+s} = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau).$$

The parity theorem makes this independent of the chosen expressions.

Definition 4.14. The *alternating group* is

$$A_n = \ker(\text{sgn}) = \{\sigma \in S_n : \sigma \text{ is even}\}.$$

It is a subgroup of S_n by the kernel property.

Example 4.15. An r-cycle has sign $(-1)^{r-1}$. In particular,

$$(1\ 2\ 3\ 4) = (1\ 4)(1\ 3)(1\ 2)$$

is odd and does not belong to A_4. A 3-cycle is even, as is a product of two disjoint transpositions.

Exercise 4.4. Fix an odd permutation $\omega \in S_n$, where $n \geq 2$. Show that every odd permutation has a unique expression $\omega\alpha$ with $\alpha \in A_n$. Deduce that $|A_n| = n!/2$.

Solution. If σ is odd, set $\alpha = \omega^{-1}\sigma$. Its sign is $(-1)(-1) = 1$, so $\alpha \in A_n$ and $\sigma = \omega\alpha$. Conversely, $\omega\alpha$ is odd whenever α is even. Cancellation gives uniqueness. Thus multiplication by ω bijects the even and odd permutations. There are $n!$ permutations, obtained by choosing their successive images in $n, n-1, \ldots, 1$ ways, so exactly half are even. For $n = 1$, $A_1 = S_1 = \{\text{id}\}$.

Chapter 5

Direct Products and Finitely Generated Abelian Groups

5.1 Products and orders of elements

Definition 5.1. For groups G and H, the *direct product* is the Cartesian product $G \times H$ with multiplication

$$(a, x)(b, y) = (ab, xy).$$

Its identity is (e_G, e_H) and $(a, x)^{-1} = (a^{-1}, x^{-1})$. More generally, a finite product $\prod_{i=1}^{k} G_i$ has componentwise multiplication.

Exercise 5.1. Verify the group axioms for a direct product, identify its factors as subgroups, and show that interchanging the factors gives an isomorphism $G \times H \cong H \times G$.

Solution. Associativity holds in each coordinate:

$$((a, x)(b, y))(c, z) = ((ab)c, (xy)z) = (a(bc), x(yz)) = (a, x)((b, y)(c, z)).$$

The identity and inverse formulas follow from the corresponding formulas in the factors. The maps $a \mapsto (a, e_H)$ and $x \mapsto (e_G, x)$ are injective homomorphisms with images $G \times \{e_H\}$ and $\{e_G\} \times H$. The map $(a, x) \mapsto (x, a)$ is a bijective homomorphism. The same coordinate argument applies to any finite number of factors. Rebracketing a finite product also preserves multiplication.

Exercise 5.2. Let $a_i \in G_i$ have finite order r_i. Prove that

$$\operatorname{ord}(a_1, \ldots, a_k) = \operatorname{lcm}(r_1, \ldots, r_k).$$

Solution. For a positive integer t,

$$(a_1, \ldots, a_k)^t = (e_1, \ldots, e_k) \quad \Longleftrightarrow \quad a_i^t = e_i \text{ for all } i \quad \Longleftrightarrow \quad r_i \mid t \text{ for all } i.$$

The least such positive t is the least common multiple. If any coordinate has infinite order, the tuple has infinite order as well.

Note 5.2. For positive integers $r_1, \ldots, r_k$, the common multiples form

$$r_1\mathbb{Z} \cap \cdots \cap r_k\mathbb{Z} = \operatorname{lcm}(r_1, \ldots, r_k)\mathbb{Z}.$$

By contrast, the subgroup generated by $r_1, \ldots, r_k$ is $\gcd(r_1, \ldots, r_k)\mathbb{Z}$.

Example 5.3. The group $\mathbb{Z}_2 \times \mathbb{Z}_2$ consists of

$$(0,0),\ (1,0),\ (0,1),\ (1,1).$$

Every nonzero element has order two, so the group is not cyclic. Identifying these elements with e, a, b, c gives the Klein four group. For $\mathbb{Z}_2 \times \mathbb{Z}_3$, the successive positive multiples of $(1,1)$ are

$$(1,1),\ (0,2),\ (1,0),\ (0,1),\ (1,2),\ (0,0).$$

Thus $(1,1)$ has order six, and $[t]_6 \mapsto ([t]_2, [t]_3)$ gives an isomorphism $\mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3$.

Theorem 5.4. *For positive integers m, n, the group $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic if and only if* $\gcd(m,n) = 1$. *In that case*

$$\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n, \qquad [t]_{mn} \longmapsto ([t]_m, [t]_n).$$

Proof. If $\gcd(m,n) = 1$, the element $(1,1)$ has order $\operatorname{lcm}(m,n) = mn$, the size of the product, so it generates the product. The displayed map is the resulting isomorphism. Conversely, put $d = \gcd(m,n)$. Every (s,t) is annihilated by mn/d, since this integer is a multiple of both m and n. If the product were generated by (s,t), its order would be mn, forcing $mn \le mn/d$ and therefore $d = 1$.

Corollary 5.5. *If $n_1, \dots, n_k$ are pairwise coprime positive integers, then*

$$\mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k} \cong \mathbb{Z}_{n_1 \cdots n_k}.$$

Proof. Induct on k. The product $n_1 \cdots n_{k-1}$ is coprime to n_k, so the preceding theorem combines the last factor with the product already combined by induction.

Exercise 5.3. Find the order of $(2,4) \in \mathbb{Z}_8 \times \mathbb{Z}_{12}$.

Solution. The coordinate orders are $8/\gcd(8,2) = 4$ and $12/\gcd(12,4) = 3$, so the tuple has order $\operatorname{lcm}(4,3) = 12$. Explicitly, $t(2,4) = (0,0)$ exactly when $4 \mid t$ and $3 \mid t$, equivalently $12 \mid t$.

5.2 Decompositions and finite generation

Definition 5.6. A nontrivial group is *decomposable* if it is isomorphic to a direct product of two nontrivial groups; otherwise it is *indecomposable*. In an internal decomposition, the factors are identified with subgroups of the original group.

Exercise 5.4. For $V = \{e, a, b, c\}$ as above, show explicitly that $V \cong \{e, a\} \times \{e, b\}$.

Solution. Let $H = \{e, a\}$ and $K = \{e, b\}$. These are subgroups, they commute elementwise, and $H \cap K = \{e\}$. The map

$$H \times K \longrightarrow V, \qquad (x, y) \longmapsto xy$$

is a homomorphism, because $(xy)(x'y') = (xx')(yy')$. It takes the four pairs to e, a, b, c, respectively, and hence is bijective. Its inverse sends each xy, with $x \in H$ and $y \in K$, to (x, y).

Definition 5.7. A group G is *finitely generated* if some finite list $a_1, \dots, a_s$ satisfies $G = \langle a_1, \dots, a_s \rangle$.

Exercise 5.5. Show that every finite group is finitely generated, that $\mathbb{Z}$ is finitely generated, and that a finite direct product of finitely generated groups is finitely generated.

Solution. A finite group is generated by the finite set of all its elements. The group $\mathbb{Z}$ is generated by 1. If $G = \langle a_1, \dots, a_r \rangle$ and $H = \langle b_1, \dots, b_s \rangle$, then $G \times H$ is generated by

$$(a_1, e_H), \dots, (a_r, e_H), \quad (e_G, b_1), \dots, (e_G, b_s).$$

Indeed, each (g, h) is $(g, e_H)(e_G, h)$, and the two factors are words in the displayed elements. Induction handles finitely many factors.

5.3 The structure theorem

Abelian groups in this section are written additively. The notation $\mathbb{Z}^r$ denotes a product of r copies of $\mathbb{Z}$, with $\mathbb{Z}^0$ the trivial group.

Exercise 5.6. Prove the following integer-basis fact. If $K \leq \mathbb{Z}^s$, there is a basis $v_1, \dots, v_s$ of $\mathbb{Z}^s$ and positive integers

$$d_1 \mid d_2 \mid \cdots \mid d_t, \qquad 0 \leq t \leq s,$$

such that $d_1 v_1, \dots, d_t v_t$ is a basis of K. Here a basis means that every element has a unique integer linear expression in the basis elements.

Solution. First prove by induction on s that every subgroup K has a finite integer basis with at most s elements. The assertion is immediate for $s = 0$. Project K onto the first coordinate. If its image is zero, use induction in the remaining $s - 1$ coordinates. Otherwise that image is $d\mathbb{Z}$ for a positive d. Choose $w \in K$ with first coordinate d. The subgroup K_0 of elements with first coordinate zero has an integer basis by induction. Every $x \in K$ has first coordinate qd and can be written uniquely as $qw + x_0$, $x_0 \in K_0$. Thus w together with a basis of K_0 is a basis of K.

Place such a basis of K into the columns of an integer matrix A with s rows and t columns. The following operations are invertible over the integers: interchange two rows or two columns, change a row or column's sign, and add an integer multiple of one row or column to another. Row operations change coordinates in the ambient integer lattice; column operations replace the chosen basis of K by another basis.

We show that these operations make A diagonal, with positive nonzero diagonal entries satisfying the divisibility condition. If A is zero there is nothing to do. Otherwise move a nonzero entry to the upper left and make it positive, calling it d. For an entry b below it, divide $b = qd + r$, $0 \leq r < d$, and subtract q times the first row from its row. If $r \neq 0$, interchange these rows to obtain a smaller positive pivot. The corresponding column procedure treats entries to the right of the pivot. Repeating the row and column procedures either decreases the positive pivot or clears its row and column. There can be only finitely many decreases. For a fixed pivot, entries divisible by it are cleared in finitely many operations without disturbing already zero entries in its column or row. We therefore reach a matrix

$$\begin{pmatrix} d & 0 \\ 0 & B \end{pmatrix}.$$

If some entry b of B is not divisible by d, add its row to the first row. The first entry remains d, while the first row now contains b. Divide b by d using a column operation and interchange columns

to obtain a positive pivot smaller than d. Repeat the clearing procedure. Again, a positive integer cannot decrease indefinitely. Eventually we obtain the displayed block form with d dividing every entry of B.

Apply the same argument inductively to B. Operations confined to B preserve the fact that all its entries are multiples of d. Its first positive diagonal entry is consequently a multiple of d, and induction gives the whole divisibility chain. The original columns were integer-linearly independent; invertible operations preserve independence, so no zero column occurs among these t columns. Interpreting the final row coordinates as an ambient basis gives precisely $K = \langle d_1 v_1, \dots, d_t v_t \rangle$, with uniqueness of these coordinates inherited from the ambient basis. This proves the fact, often called Smith normal form for an integer sublattice.

Theorem 5.8 (Fundamental theorem of finitely generated abelian groups)**.** *Every finitely generated abelian group has an invariant-factor decomposition*

$$G \cong \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_t} \times \mathbb{Z}^r, \qquad 2 \le d_1 \mid d_2 \mid \cdots \mid d_t.$$

The integer r and the sequence $d_1, \dots, d_t$ are unique. Equivalently, it has a primary decomposition

$$G \cong \mathbb{Z}_{p_1^{e_1}} \times \cdots \times \mathbb{Z}_{p_\ell^{e_\ell}} \times \mathbb{Z}^r,$$

where each p_i is prime and each $e_i \ge 1$. The prime-power factors are unique up to reordering; primes may repeat. The integer r is the rank*, also called the* Betti number*. The integers d_i are the* invariant factors*.*

Exercise 5.7. Prove existence of both decompositions in the fundamental theorem.

Solution. Choose generators $a_1, \dots, a_s$ and define

$$F : \mathbb{Z}^s \to G, \qquad F(n_1, \dots, n_s) = n_1 a_1 + \cdots + n_s a_s.$$

This is a surjective homomorphism. Its relation subgroup is $K = \{x \in \mathbb{Z}^s : F(x) = 0\}$. Apply the integer-basis fact: for an ambient basis $v_1, \dots, v_s$, the relations have basis $d_1 v_1, \dots, d_t v_t$, with $d_1 \mid \cdots \mid d_t$. Put $b_i = F(v_i)$. Every element of G is an integer combination of the b_i, and

$$\sum_{i=1}^{s} n_i b_i = 0 \quad \Longleftrightarrow \quad d_i \mid n_i \ (1 \le i \le t), \quad n_i = 0 \ (i > t).$$

Thus the coordinate map

$$\mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_t} \times \mathbb{Z}^{s-t} \longrightarrow G, \quad ([n_1]_{d_1}, \dots, [n_t]_{d_t}, n_{t+1}, \dots, n_s) \longmapsto \sum_i n_i b_i$$

is well defined, a homomorphism, surjective, and injective by this exact description of the relations. Factors with $d_i = 1$ are trivial and may be removed. This proves invariant-factor existence.

Factor each remaining d_i into powers of distinct primes. Those prime powers are pairwise coprime, so the cyclic-product theorem splits $\mathbb{Z}_{d_i}$ into the product of the associated prime-power cyclic groups. Reordering all factors gives the primary decomposition.

Exercise 5.8. Prove the uniqueness assertions in the fundamental theorem.

Solution. The set

$$T = \{x \in G : nx = 0 \text{ for some positive integer } n\}$$

is preserved by every isomorphism. It is a subgroup: if $nx = 0$ and $my = 0$, then $nm(x - y) = 0$. In a displayed decomposition $G \cong \mathbb{Z}^r \times T_0$, with T_0 finite, it is exactly $\{0\} \times T_0$. In particular it is finite.

The rank r can be recovered as the largest size of an integer-linearly independent list of elements of G. The r standard infinite coordinates are independent. If $r + 1$ elements are given, their projections to $\mathbb{Z}^r$ are linearly dependent over $\mathbb{Q}$. Clearing denominators gives an integer relation with coefficients not all zero whose value in G lies in T. Multiplying all coefficients by any positive integer annihilating T makes that value zero. Such an annihilator exists, for example the product of the orders of all elements of the finite group T. Thus no $r + 1$ elements are independent, and r is determined by G.

To recover the finite factors, for each prime p and $k \geq 0$ let

$$T[p^k] = \{x \in T : p^k x = 0\}, \qquad |T[p^k]| = p^{a_p(k)}.$$

In a cyclic group of order p^e, exactly $p^{\min(k,e)}$ elements are annihilated by p^k: the condition on $[x]_{p^e}$ is $p^e \mid p^k x$. In a cyclic group of order q^e with $q \neq p$, only zero is annihilated by p^k, since p^k is coprime to q^e. Consequently, if the p-power factors have exponents $e_1, \ldots, e_u$, then

$$a_p(k) = \sum_{j=1}^{u} \min(k, e_j), \qquad a_p(k) - a_p(k-1) = \#\{j : e_j \geq k\} \quad (k \geq 1).$$

The cardinalities $|T[p^k]|$ are intrinsic, so these differences determine the number of exponents at least k, hence the number equal to each k. This recovers every prime-power factor with multiplicity and proves primary uniqueness.

Finally, for each prime arrange its exponents in nondecreasing order and pad the list on the left with zeros so that all lists have a common length equal to the largest number of factors for one prime. Multiply the prime powers in corresponding positions. The resulting integers form a divisibility chain $d_1 \mid \cdots \mid d_t$, all greater than one. In any invariant-factor decomposition, divisibility forces the exponent lists for every prime to be aligned in exactly this way: all zero exponents precede positive ones. Therefore primary uniqueness also determines the invariant factors uniquely.

5.4 Classification examples and consequences

Exercise 5.9. List all finite abelian groups of order 72, up to isomorphism, in primary form and invariant-factor form.

Solution. Since $72 = 2^3 3^2$, the 2-power part has one of the three forms

$$\mathbb{Z}_8, \qquad \mathbb{Z}_2 \times \mathbb{Z}_4, \qquad \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2,$$

and the 3-power part has one of the two forms $\mathbb{Z}_9$ or $\mathbb{Z}_3 \times \mathbb{Z}_3$. Taking all six pairs gives the complete

list. Combining coprime factors in aligned columns gives

Primary form	Invariant-factor form
$\mathbb{Z}_8 \times \mathbb{Z}_9$	$\mathbb{Z}_{72}$
$\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_9$	$\mathbb{Z}_2 \times \mathbb{Z}_{36}$
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9$	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{18}$
$\mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_3$	$\mathbb{Z}_3 \times \mathbb{Z}_{24}$
$\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3$	$\mathbb{Z}_6 \times \mathbb{Z}_{12}$
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3$	$\mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_6$

The partitions 3, 2+1, 1+1+1 and 2, 1+1 exhaust the possible prime-power exponents. Uniqueness in the structure theorem shows that these six groups are pairwise nonisomorphic.

Corollary 5.9. *A nontrivial finite abelian group is indecomposable if and only if it is isomorphic to $\mathbb{Z}_{p^e}$ for some prime p and positive integer e. In particular, $\mathbb{Z}_5$ and $\mathbb{Z}_{25}$ are indecomposable.*

Exercise 5.10. Prove the corollary.

Solution. If the primary decomposition has at least two factors, grouping one factor against the rest gives a product of two nontrivial groups. Thus an indecomposable group has just one prime-power cyclic factor. Conversely, suppose $\mathbb{Z}_{p^e} \cong H \times K$ with both factors nontrivial. A projection of a cyclic group is cyclic, so H and K are cyclic. Their cardinalities multiply to p^e, so write $|H| = p^a$ and $|K| = p^b$ with $a, b \geq 1$ and $a + b = e$. The cyclic-product theorem shows that their product is not cyclic, because $\gcd(p^a, p^b) > 1$. This contradicts the original isomorphism.

Chapter 6

Cosets, Quotients, and Normal Subgroups

6.1 Cosets and equivalence relations

Definition 6.1. Let $H \leq G$. For $a \in G$, the *left coset* and *right coset* of H represented by a are

$$aH = \{ah : h \in H\}, \qquad Ha = \{ha : h \in H\}.$$

The set of left cosets is denoted by G/H. At this stage, this notation describes a set; a group operation on it requires an additional hypothesis.

Exercise 6.1. For $a, b \in G$, define $a \sim_L b$ by $a^{-1}b \in H$. Prove that this is an equivalence relation and that the equivalence class of a is aH. Show also that

$$aH = bH \iff a^{-1}b \in H \iff b = ah \text{ for some } h \in H.$$

Solution. Reflexivity follows from $a^{-1}a = e \in H$. If $a^{-1}b \in H$, then its inverse $b^{-1}a$ belongs to H, proving symmetry. If $a^{-1}b, b^{-1}c \in H$, their product $a^{-1}c$ belongs to H, proving transitivity. Moreover,

$$\{b : a \sim_L b\} = \{b : a^{-1}b \in H\} = \{ah : h \in H\} = aH.$$

Thus these cosets partition G. If $b = ah$ with $h \in H$, then $bH = ahH = aH$, since $hH = H$. Conversely, $aH = bH$ implies $b \in aH$, giving the required expression for b.

Proposition 6.2. *Two left cosets of a subgroup are either equal or disjoint, and every element of G belongs to exactly one left coset.*

Proof. Every $a \in G$ belongs to aH. If $x \in aH \cap bH$, write $x = ah = bk$ with $h, k \in H$. Then $a^{-1}b = hk^{-1} \in H$, so $aH = bH$. Consequently distinct left cosets are disjoint.

Example 6.3. In the additive group $\mathbb{Z}$, the subgroup $4\mathbb{Z}$ has exactly four cosets:

$$4\mathbb{Z}, \quad 1 + 4\mathbb{Z}, \quad 2 + 4\mathbb{Z}, \quad 3 + 4\mathbb{Z}.$$

The division algorithm writes each integer uniquely as $4q + r$ with $0 \leq r < 4$, so these sets form a disjoint partition of $\mathbb{Z}$.

Example 6.4. In the additive group $\mathbb{Z}_{12}$, let $H = \langle 3 \rangle$. Writing residues by their standard representatives,

$$H = \{0, 3, 6, 9\}, \qquad 1 + H = \{1, 4, 7, 10\}, \qquad 2 + H = \{2, 5, 8, 11\}.$$

These three cosets partition $\mathbb{Z}_{12}$. In particular, $|H| = 4$ and there are three left cosets.

Exercise 6.2. Define $a \sim_R b$ by $ab^{-1} \in H$. Show that this is an equivalence relation whose classes are the right cosets Ha. Prove that Ha, H, and aH have the same cardinality, and that the sets of left and right cosets have the same cardinality.

Solution. We have $aa^{-1} = e \in H$. If $ab^{-1} \in H$, then $ba^{-1} = (ab^{-1})^{-1} \in H$. If $ab^{-1}, bc^{-1} \in H$, then their product ac^{-1} belongs to H. Hence $\sim_R$ is an equivalence relation. The condition $ab^{-1} \in H$ is equivalent to $b \in Ha$, so its classes are right cosets. The maps

$$H \longrightarrow aH, \quad h \longmapsto ah, \qquad H \longrightarrow Ha, \quad h \longmapsto ha$$

are bijections, with inverse maps $x \mapsto a^{-1}x$ and $x \mapsto xa^{-1}$ respectively. Finally, inversion sends the set aH to Ha^{-1}. Thus

$$aH \longmapsto Ha^{-1}$$

is a well-defined bijection between the sets of left and right cosets; its inverse is $Hb \mapsto b^{-1}H$.

6.2 Lagrange's theorem and index

Definition 6.5. The *index* of H in G, written $[G : H]$, is the number of left cosets of H in G. This may be an infinite cardinal. Equivalently, it is the number of right cosets.

Lemma 6.6. *Every left coset aH has the same cardinality as H.*

Proof. The map $H \to aH$ given by $h \mapsto ah$ is surjective by the definition of a coset and injective by cancellation.

Theorem 6.7 (Lagrange's theorem)**.** *If G is a finite group and $H \leq G$, then*

$$|G| = [G : H]|H|.$$

In particular, $|H|$ divides $|G|$.

Proof. The left cosets of H partition G, and each coset contains exactly $|H|$ elements. Adding the sizes of these $[G : H]$ disjoint sets gives the formula.

Example 6.8. The coset calculations above give

$$[\mathbb{Z} : 4\mathbb{Z}] = 4, \qquad [\mathbb{Z}_{12} : \langle 3 \rangle] = 3.$$

The first index is finite even though the group and subgroup are infinite. The second also follows from Lagrange's formula, since $12/4 = 3$.

Exercise 6.3. Prove that the order of an element of a finite group divides the order of the group. Deduce that every group of prime order is cyclic.

Solution. For $a \in G$, the cyclic subgroup $\langle a \rangle$ has order $\operatorname{ord}(a)$, so Lagrange's theorem gives $\operatorname{ord}(a) \mid |G|$. Now suppose $|G| = p$ is prime. Choose $a \neq e$. Then $\operatorname{ord}(a) > 1$ and divides p, so $\operatorname{ord}(a) = p$. Therefore $\langle a \rangle = G$.

Exercise 6.4. Let $H \leq K \leq G$, with G finite. Prove the multiplicativity of index:

$$[G : H] = [G : K][K : H].$$

Solution. Lagrange's theorem gives

$$[G : K][K : H] = \frac{|G|}{|K|}\frac{|K|}{|H|} = \frac{|G|}{|H|} = [G : H].$$

One can also see the partition directly. Choose representatives $a_1, \ldots, a_r$ for the left cosets of K in G, and $b_1, \ldots, b_s$ for those of H in K. Every element of G lies in some $a_i b_j H$. If $a_i b_j H = a_{i'} b_{j'} H$, then $a_i K = a_{i'} K$, so $i = i'$, and cancellation then gives $b_j H = b_{j'} H$, so $j = j'$. Thus the rs products represent exactly the distinct cosets of H.

Exercise 6.5. Use the structure theorem for finite abelian groups to show that if G is finite abelian and m is a positive divisor of $|G|$, then G has a subgroup of order m.

Solution. Write the prime-power decomposition supplied by the structure theorem as

$$G \cong \prod_p \prod_{i=1}^{r_p} \mathbb{Z}_{p^{a_{p,i}}}.$$

Only finitely many factors occur. Write $m = \prod_p p^{b_p}$, where $0 \leq b_p \leq \sum_i a_{p,i}$. For each p, choose integers $c_{p,i}$ satisfying

$$0 \leq c_{p,i} \leq a_{p,i}, \qquad \sum_i c_{p,i} = b_p.$$

Such a choice is obtained by filling the first factor up to its capacity $a_{p,1}$, then the second, and continuing until the total is b_p. In $\mathbb{Z}_{p^{a_{p,i}}}$ the element $p^{a_{p,i}-c_{p,i}}$ generates a subgroup of order $p^{c_{p,i}}$, with the trivial subgroup used when $c_{p,i} = 0$. The product of these subgroups has order

$$\prod_p \prod_i p^{c_{p,i}} = \prod_p p^{b_p} = m.$$

Its inverse image under the displayed isomorphism is the desired subgroup of G.

6.3 Fibers of homomorphisms

Exercise 6.6. Let $\varphi : G \to G'$ be a homomorphism and put $K = \ker \varphi$. Prove that

$$\varphi(a) = \varphi(b) \quad \Longleftrightarrow \quad aK = bK.$$

Identify the fiber over $\varphi(a)$.

Solution. The kernel is a subgroup by the homomorphism properties already proved. For $a, b \in G$,

$$\varphi(a) = \varphi(b) \quad \Longleftrightarrow \quad \varphi(a^{-1}b) = e' \quad \Longleftrightarrow \quad a^{-1}b \in K \quad \Longleftrightarrow \quad aK = bK.$$

Thus the entire fiber over $\varphi(a)$ is the left coset aK. In particular, this also recovers the injectivity criterion: all fibers are singletons exactly when $K = \{e\}$.

Example 6.9 (Determinants and special linear groups)**.** For $n \geq 1$, let $\mathrm{GL}_n(\mathbb{R})$ be the group of invertible real $n \times n$ matrices, and let $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$ under multiplication. Since

$$\det(AB) = \det(A)\det(B),$$

the determinant defines a homomorphism

$$\det : \mathrm{GL}_n(\mathbb{R}) \longrightarrow \mathbb{R}^\times.$$

Its kernel is the *special linear group*

$$\mathrm{SL}_n(\mathbb{R}) = \{A \in \mathrm{GL}_n(\mathbb{R}) : \det A = 1\}.$$

Thus for invertible matrices A, B,

$$A\,\mathrm{SL}_n(\mathbb{R}) = B\,\mathrm{SL}_n(\mathbb{R}) \quad \Longleftrightarrow \quad \det A = \det B.$$

6.4 Normal subgroups and quotient groups

Definition 6.10. A subgroup $H \leq G$ is *normal*, written $H \trianglelefteq G$, if

$$aH = Ha \qquad \text{for every } a \in G.$$

Exercise 6.7. For a subgroup $H \leq G$, prove that the following conditions are equivalent:

1. $H \trianglelefteq G$;
2. $aHa^{-1} = H$ for every $a \in G$;
3. $aha^{-1} \in H$ for every $a \in G$ and $h \in H$.

Solution. Multiplying the equality $aH = Ha$ on the right by a^{-1} gives $aHa^{-1} = H$, and reversing this multiplication proves the converse. The equality $aHa^{-1} = H$ immediately implies that each aha^{-1} lies in H. Now assume the third condition. It gives $aHa^{-1} \subseteq H$. Applying the same condition with a^{-1} in place of a gives $a^{-1}Ha \subseteq H$. Multiplying this inclusion on the left by a and on the right by a^{-1} yields $H \subseteq aHa^{-1}$. Hence equality holds. Equivalently, the two coset inclusions can be checked directly: $ah = (aha^{-1})a \in Ha$, and $ha = a(a^{-1}ha) \in aH$. Therefore $aH = Ha$.

Example 6.11. Every subgroup of an abelian group is normal, since $ah = ha$. In particular, $5\mathbb{Z} \trianglelefteq \mathbb{Z}$.

Example 6.12. Let

$$H = A_3 = \{e, (1\,2\,3), (1\,3\,2)\} \leq S_3.$$

Conjugation sends a cycle $(i\,j\,k)$ to $(\sigma(i)\,\sigma(j)\,\sigma(k))$. Thus conjugating a 3-cycle in S_3 produces one of the two 3-cycles, and $\sigma H \sigma^{-1} = H$ for every $\sigma \in S_3$. Hence $A_3 \trianglelefteq S_3$. For example, using right-to-left composition,

$$(1\,2)H = \{(1\,2), (2\,3), (1\,3)\}, \qquad H(1\,2) = \{(1\,2), (1\,3), (2\,3)\}.$$

The listings differ in order but describe the same coset.

Example 6.13. The subgroup $\mathrm{SL}_n(\mathbb{R})$ is normal in $\mathrm{GL}_n(\mathbb{R})$. Indeed, if $A \in \mathrm{SL}_n(\mathbb{R})$ and $B \in \mathrm{GL}_n(\mathbb{R})$, then

$$\det(BAB^{-1}) = \det(B)\det(A)\det(B)^{-1} = 1,$$

so $BAB^{-1} \in \mathrm{SL}_n(\mathbb{R})$.

Theorem 6.14. *Let $H \trianglelefteq G$. Multiplication of cosets by*

$$(aH)(bH) = (ab)H$$

is well-defined.

Proof. Suppose $aH = xH$ and $bH = yH$. Then $x = ah$ and $y = bk$ for some $h, k \in H$. Consequently,

$$xy = ahbk = ab(b^{-1}hb)k.$$

Normality gives $b^{-1}hb \in H$, and therefore $(b^{-1}hb)k \in H$. It follows that $xy \in abH$, so $(xy)H = (ab)H$. Thus the resulting coset is independent of the chosen representatives.

Exercise 6.8. Verify that this multiplication makes G/H a group. Conversely, if the rule $(aH)(bH) = (ab)H$ is well-defined on the left cosets of a subgroup $H \leq G$, prove that H is normal.

Solution. Assume first that H is normal. The product is a coset, so closure holds. Associativity follows from that in G:

$$((aH)(bH))(cH) = ((ab)c)H = (a(bc))H = (aH)((bH)(cH)).$$

The identity is $H = eH$, since $(aH)H = H(aH) = aH$. The inverse of aH is $a^{-1}H$, because both products are H. Hence all group axioms hold. For the converse, suppose the rule is well-defined. For $h \in H$ we have $hH = H$, so for any $a \in G$,

$$(ha)H = (hH)(aH) = H(aH) = aH.$$

Equality of these cosets gives $a^{-1}ha \in H$. As a^{-1} runs through all elements of G, this is the conjugation criterion for normality.

Definition 6.15. If $H \trianglelefteq G$, the group of cosets with this multiplication is the *quotient group* G/H. The map

$$\pi : G \longrightarrow G/H, \qquad \pi(a) = aH,$$

is the *canonical projection*. It is a surjective homomorphism and $\ker \pi = H$, since $aH = H$ exactly when $a \in H$.

Example 6.16. The quotient $\mathbb{Z}/5\mathbb{Z}$ has elements

$$5\mathbb{Z},\ 1+5\mathbb{Z},\ 2+5\mathbb{Z},\ 3+5\mathbb{Z},\ 4+5\mathbb{Z},$$

and its operation is

$$(a+5\mathbb{Z}) + (b+5\mathbb{Z}) = (a+b) + 5\mathbb{Z}.$$

The map $a + 5\mathbb{Z} \mapsto [a]_5$ identifies this group with $\mathbb{Z}_5$. More generally, for every positive integer n, reduction modulo n is a surjective homomorphism

$$\rho_n : \mathbb{Z} \longrightarrow \mathbb{Z}_n, \qquad a \longmapsto [a]_n,$$

with kernel $n\mathbb{Z}$. Equality $a + n\mathbb{Z} = b + n\mathbb{Z}$ is equivalent to $n \mid a - b$, so

$$\mathbb{Z}/n\mathbb{Z} \longrightarrow \mathbb{Z}_n, \qquad a + n\mathbb{Z} \longmapsto [a]_n$$

is well-defined and bijective. The addition formula for cosets makes it an isomorphism.

6.5 The first isomorphism theorem

Exercise 6.9 (First isomorphism theorem)**.** Let $\varphi : G \to G'$ be a group homomorphism. Prove that $\ker\varphi$ is normal and that

$$\overline{\varphi} : G/\ker\varphi \longrightarrow \operatorname{im}\varphi, \qquad a\ker\varphi \longmapsto \varphi(a)$$

is an isomorphism. Show that $\varphi = \iota \circ \overline{\varphi} \circ \pi$, where π is the canonical projection and $\iota : \operatorname{im}\varphi \hookrightarrow G'$ is inclusion.

Solution. Put $K = \ker\varphi$. If $k \in K$ and $a \in G$, then

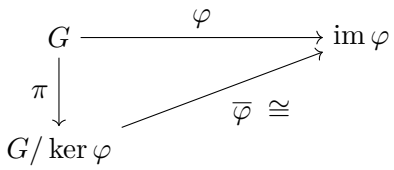
$$\varphi(aka^{-1}) = \varphi(a)e'\varphi(a)^{-1} = e',$$

so $aka^{-1} \in K$. Thus $K \trianglelefteq G$. The fiber criterion shows that $aK = bK$ exactly when $\varphi(a) = \varphi(b)$. Its forward implication makes $\overline{\varphi}$ well-defined; its reverse implication makes $\overline{\varphi}$ injective. The map is surjective onto $\operatorname{im}\varphi$ by the definition of image. Finally,

$$\overline{\varphi}((aK)(bK)) = \overline{\varphi}(abK) = \varphi(ab) = \varphi(a)\varphi(b) = \overline{\varphi}(aK)\overline{\varphi}(bK).$$

Hence it is an isomorphism. For every $a \in G$, the composite $\iota\overline{\varphi}\pi$ sends a to $\varphi(a)$, proving the asserted factorization. In particular, if φ is surjective, then $G/\ker\varphi \cong G'$.

The factorization can be displayed by the commutative diagram



where the top arrow regards φ as a map onto its image.

Example 6.17. The determinant map is onto $\mathbb{R}^\times$: for $t \neq 0$, the matrix $\operatorname{diag}(t, 1, \ldots, 1)$ has determinant t. Therefore

$$\mathrm{GL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{R}) \cong \mathbb{R}^\times, \qquad A\,\mathrm{SL}_n(\mathbb{R}) \longmapsto \det A.$$

6.6 Normal images, inverse images, and simple quotients

Exercise 6.10. Let $\varphi : G \to G'$ be a homomorphism. Prove the following assertions:

1. If $N \trianglelefteq G$, then $\varphi(N) \trianglelefteq \varphi(G)$.
2. If $N' \trianglelefteq G'$, then $\varphi^{-1}(N') \trianglelefteq G$.

Solution. The image $\varphi(N)$ contains the identity and is closed under products and inverses, so it is a subgroup of $\varphi(G)$. For $g \in G$ and $n \in N$,

$$\varphi(g)\varphi(n)\varphi(g)^{-1} = \varphi(gng^{-1}) \in \varphi(N),$$

since N is normal. This proves the first assertion. Normality is asserted in $\varphi(G)$; if φ is onto, this is all of G'. For the second assertion, put $L = \varphi^{-1}(N')$. The identity belongs to L, and for $x, y \in L$,

$$\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} \in N',$$

so L is a subgroup. If $g \in G$ and $x \in L$, normality of N' gives

$$\varphi(gxg^{-1}) = \varphi(g)\varphi(x)\varphi(g)^{-1} \in N'.$$

Thus $gxg^{-1} \in L$, proving $L \trianglelefteq G$.

Definition 6.18. A group G is *simple* if $G \neq \{e\}$ and its only normal subgroups are $\{e\}$ and G. A *maximal normal subgroup* is a proper normal subgroup $M \trianglelefteq G$ for which there is no normal subgroup $L \trianglelefteq G$ satisfying $M \subsetneq L \subsetneq G$.

Example 6.19. The cyclic group $\mathbb{Z}_4$ is not simple, since

$$\langle 2 \rangle = \{0, 2\}$$

is a nontrivial proper subgroup and is normal because $\mathbb{Z}_4$ is abelian.

Exercise 6.11. Prove that $\mathbb{Z}_p$ is simple for every prime p.

Solution. Every subgroup of $\mathbb{Z}_p$ has order dividing p, hence has order 1 or p. Therefore there is no nontrivial proper subgroup, and in particular no nontrivial proper normal subgroup. Since $p > 1$, the group is nontrivial and is simple.

Exercise 6.12 (Maximal normal subgroups and simple quotients)**.** Let $M \trianglelefteq G$ with $M \neq G$. Prove that M is maximal normal if and only if G/M is simple.

Solution. Let $\pi : G \to G/M$ be the canonical projection. For any subgroup $N \leq G/M$, its inverse image contains M and satisfies

$$\pi(\pi^{-1}(N)) = N,$$

because π is onto. If $L \leq G$ contains M, then

$$\pi^{-1}(\pi(L)) = L.$$

Indeed, if $g \in \pi^{-1}(\pi(L))$, there is $\ell \in L$ with $gM = \ell M$. Hence $g = \ell m$ for some $m \in M \subseteq L$, and $g \in L$; the reverse inclusion is immediate. Suppose M is maximal normal and $N \trianglelefteq G/M$. Then $\pi^{-1}(N)$ is normal in G and contains M, so it is M or G. Taking its image shows that N is $\{M\}$ or G/M. The quotient is nontrivial because $M \neq G$, so it is simple. Conversely, suppose G/M is simple and $M \subseteq L \trianglelefteq G$. The normal image property gives $\pi(L) \trianglelefteq G/M$, hence $\pi(L) = \{M\}$ or G/M. Taking inverse images gives $L = M$ or G. Thus M is maximal normal.

Exercise 6.13. In $G = \mathbb{Z}_4 \times \mathbb{Z}_6$, compute the quotients by

$$H = \langle (1,1) \rangle, \qquad K = \langle (1,2) \rangle,$$

and determine whether these are maximal normal subgroups.

Solution. The order of $(1,1)$ is $\operatorname{lcm}(4,6) = 12$, and the order of $(1,2)$ is $\operatorname{lcm}(4,3) = 12$. Since $|G| = 24$, both subgroups have index 2. For an explicit quotient map, define

$$f : G \longrightarrow \mathbb{Z}_2, \qquad f(a,b) = [a-b]_2.$$

This is well-defined because changing a by 4 or b by 6 does not change its parity, and it is an additive homomorphism. It is onto, since $f(1,0) = 1$. Every multiple (k,k) of $(1,1)$ lies in its kernel. The kernel consists of pairs of equal parity: there are two choices of a and three of b for each parity, giving 12 elements. Hence $\ker f = H$, and the first isomorphism theorem yields $G/H \cong \mathbb{Z}_2$. Similarly,

$$g : G \longrightarrow \mathbb{Z}_2, \qquad g(a,b) = [b]_2$$

is a well-defined surjective homomorphism. Its kernel consists of the $4 \cdot 3 = 12$ pairs with even second coordinate. Every multiple of $(1,2)$ lies in this kernel, so $\ker g = K$ and $G/K \cong \mathbb{Z}_2$. Both H and K are normal because G is abelian. Their quotients are simple, so both are maximal normal. They are distinct: $(1,1) \notin K$ and $(1,2) \notin H$.

6.7 Simplicity of alternating groups

Exercise 6.14. Prove that A_n is simple for every $n \geq 5$.

Solution. We use right-to-left composition. First, A_n is generated by 3-cycles. An even permutation is a product of an even number of transpositions; group the transpositions into pairs. Identical transpositions cancel, a pair sharing one letter is a 3-cycle, and a pair of disjoint transpositions satisfies

$$(a\,b)(c\,d) = (a\,c\,b)(a\,c\,d).$$

Thus every even permutation is a product of 3-cycles. All 3-cycles are conjugate within A_n. Given two 3-cycles α, β, choose $s \in S_n$ carrying the ordered letters of α to those of β. Then $s\alpha s^{-1} = \beta$. If s is odd, choose a transposition t of two letters outside the support of α, which is possible since $n \geq 5$. This t commutes with α, so st is even and still conjugates α to β. Now let $N \trianglelefteq A_n$ be nontrivial. For a permutation σ, its support is

$$\operatorname{supp}(\sigma) = \{i : \sigma(i) \neq i\}.$$

Choose a nonidentity $\sigma \in N$ with support of least possible size m. We show that $m = 3$. Choose a with $\sigma(a) = b \neq a$, and choose distinct c, d outside $\{a, b\}$. Put $\tau = (a\,c\,d)$. The conjugate $\sigma\tau\sigma^{-1}$ has b in its support, whereas τ does not. Thus they are different, and

$$\delta = \sigma\tau\sigma^{-1}\tau^{-1}$$

is nonidentity. It lies in N, since $\sigma \in N$ and $\tau\sigma^{-1}\tau^{-1} \in N$. Its support is contained in $\operatorname{supp}(\tau) \cup \sigma(\operatorname{supp}(\tau))$, a set of at most six letters. Minimality therefore gives $m \leq 6$. A nonidentity even permutation cannot move just one or two letters, so $3 \leq m \leq 6$. If $m = 3$, then σ is a 3-cycle, as desired. The other possibilities are ruled out as follows. If $m = 4$, parity forces $\sigma = (a\,b)(c\,d)$. Choose a fifth letter e and put $\tau = (c\,d\,e)$. Then

$$\sigma\tau\sigma^{-1} = (d\,c\,e) = \tau^{-1}, \qquad \sigma\tau\sigma^{-1}\tau^{-1} = \tau^{-2} = \tau.$$

This is a 3-cycle in N, contradicting minimality of $m = 4$. If $m = 5$, parity forces $\sigma = (a\,b\,c\,d\,e)$. With $\tau = (a\,b\,c)$ we obtain

$$\sigma\tau\sigma^{-1}\tau^{-1} = (b\,c\,d)(a\,c\,b) = (a\,d\,b),$$

a 3-cycle in N, again contradicting minimality. If $m = 6$, the disjoint-cycle types of an even permutation moving all six letters are (4)(2) and (3)(3). In the first case, write $\sigma = (a\,b\,c\,d)(e\,f)$ and take $\tau = (a\,b\,c)$. The same calculation gives $(a\,d\,b) \in N$. In the second case, write $\sigma = (a\,b\,c)(d\,e\,f)$ and take $\tau = (a\,b\,d)$. Then

$$\sigma\tau\sigma^{-1}\tau^{-1} = (b\,c\,e)(a\,d\,b).$$

This element moves a to d and has support contained in $\{a, b, c, d, e\}$. It is a nonidentity element of N moving at most five letters, contradicting the minimal choice $m = 6$. Consequently N contains a 3-cycle. Since N is normal and all 3-cycles are conjugate in A_n, it contains every 3-cycle. These generate A_n, so $N = A_n$. The group A_n is nontrivial, and therefore simple.

6.8 Centers, commutators, and abelianization

Definition 6.20. The *center* of a group G is

$$Z(G) = \{z \in G : zg = gz \text{ for every } g \in G\}.$$

For $a, b \in G$, their *commutator* is

$$[a, b] = aba^{-1}b^{-1}.$$

The *commutator subgroup* is the subgroup generated by all such elements:

$$[G, G] = \langle [a, b] : a, b \in G \rangle.$$

Example 6.21. If G is abelian, every element commutes with every other element, so $Z(G) = G$. Conversely, $Z(G) = G$ implies that G is abelian.

Exercise 6.15. Compute the center of $\mathrm{GL}_2(\mathbb{R})$.

Solution. Every nonzero scalar matrix commutes with all matrices, so

$$\{\lambda I_2 : \lambda \in \mathbb{R}^\times\} \subseteq Z(\mathrm{GL}_2(\mathbb{R})).$$

For the opposite inclusion, let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ belong to the center. It commutes with the invertible matrix $D = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$, so

$$AD = \begin{pmatrix} a & 2b \\ c & 2d \end{pmatrix} = \begin{pmatrix} a & b \\ 2c & 2d \end{pmatrix} = DA.$$

Hence $b = c = 0$. It also commutes with $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and therefore

$$AP = \begin{pmatrix} 0 & a \\ d & 0 \end{pmatrix} = \begin{pmatrix} 0 & d \\ a & 0 \end{pmatrix} = PA,$$

which gives $a = d$. Since A is invertible, $a \neq 0$. Thus

$$Z(\mathrm{GL}_2(\mathbb{R})) = \left\{ \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} : \lambda \in \mathbb{R}^\times \right\}.$$

Exercise 6.16. Prove that $Z(G)$ and $[G, G]$ are normal subgroups of G.

Solution. The identity is central. If $z, w \in Z(G)$ and $g \in G$, then

$$(zw^{-1})g = z(w^{-1}g) = z(gw^{-1}) = (gz)w^{-1} = g(zw^{-1}),$$

so $zw^{-1} \in Z(G)$ and $Z(G) \leq G$. For $z \in Z(G)$ and $g \in G$, $gzg^{-1} = z$, so the center is normal. By definition $[G, G]$ is already a subgroup. Conjugation carries a generating commutator to another commutator:

$$g[a, b]g^{-1} = (gag^{-1})(gbg^{-1})(gag^{-1})^{-1}(gbg^{-1})^{-1} = [gag^{-1}, gbg^{-1}].$$

Every element of $[G, G]$ is a finite product of commutators and their inverses. Conjugation preserves products and inverses and sends every generator back into $[G, G]$. Hence $g[G, G]g^{-1} \subseteq [G, G]$ for every $g \in G$, which proves normality.

Definition 6.22. The quotient $G/[G, G]$ is called the *abelianization* of G.

Exercise 6.17. Show that the abelianization is abelian and that, for a normal subgroup $N \trianglelefteq G$,

$$G/N \text{ is abelian} \quad \iff \quad [G, G] \subseteq N.$$

Solution. For $a, b \in G$, the cosets aN, bN commute exactly when

$$(aN)(bN)(aN)^{-1}(bN)^{-1} = N,$$

which is equivalent to $[a, b] \in N$. Thus G/N is abelian exactly when N contains every commutator. Since N is a subgroup, this is equivalent to $[G, G] \subseteq N$. Taking $N = [G, G]$ proves that the abelianization is abelian.

Chapter 7

The First Examples of Rings

7.1 Ring axioms, unity, and commutativity

Definition 7.1. A *ring* is a set R equipped with two binary operations, addition and multiplication, such that:

1. $(R, +)$ is an abelian group, with additive identity denoted by 0;
2. multiplication is associative: $(ab)c = a(bc)$ for all $a, b, c \in R$;
3. both distributive laws hold:

$$a(b + c) = ab + ac, \qquad (a + b)c = ac + bc \quad (a, b, c \in R).$$

An element $1 \in R$ satisfying $1a = a1 = a$ for every $a \in R$ is a *unity* or *multiplicative identity*. A ring is *commutative* if $ab = ba$ for all $a, b \in R$. A ring need not have a unity under this definition.

Definition 7.2. A *field* is a commutative ring with a unity $1 \neq 0$ in which each nonzero element a has a multiplicative inverse: there is b with $ab = ba = 1$.

Example 7.3. With their usual addition and multiplication, $\mathbb{Z}$ and $\mathbb{R}$ are commutative rings with unity 1. Associativity and distributivity are the usual arithmetic laws, and each is an abelian group under addition. Every nonzero real number a has inverse $1/a$, so $\mathbb{R}$ is a field. The ring $\mathbb{Z}$ is not a field, since $2b = 1$ has no solution $b \in \mathbb{Z}$.

7.2 Matrices, polynomials, and even integers

Exercise 7.1. Verify that $M_2(\mathbb{R})$, the set of all real 2×2 matrices with matrix addition and multiplication, is a ring with unity. Determine whether it is commutative.

Solution. Addition is entrywise, so the additive laws follow from those in $\mathbb{R}$. The zero matrix is its identity and $-A$ is the additive inverse of A. For matrices $A = (a_{ij})$, $B = (b_{ij})$, $C = (c_{ij})$, every entry of AB is the real number $\sum_{k=1}^{2} a_{ik}b_{kj}$, so multiplication is closed. Associativity follows entrywise from

$$((AB)C)_{ij} = \sum_{\ell=1}^{2} \sum_{k=1}^{2} a_{ik}b_{k\ell}c_{\ell j} = \sum_{k=1}^{2} a_{ik} \sum_{\ell=1}^{2} b_{k\ell}c_{\ell j} = (A(BC))_{ij}.$$

The two distributive laws follow from

$$(A(B+C))_{ij} = \sum_{k=1}^{2} a_{ik}(b_{kj}+c_{kj}) = (AB)_{ij} + (AC)_{ij},$$
$$((A+B)C)_{ij} = \sum_{k=1}^{2} (a_{ik}+b_{ik})c_{kj} = (AC)_{ij} + (BC)_{ij}.$$

The matrix

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

satisfies $I_2 A = A I_2 = A$ and is the unity. To test commutativity, take

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

They are unequal, so this ring is not commutative.

Example 7.4. The polynomials with integer coefficients form the commutative ring $\mathbb{Z}[x]$ under coefficientwise addition and ordinary polynomial multiplication. Its unity is the constant polynomial 1. The general construction and a full verification for $R[x]$ are given in Section 8.2.

Exercise 7.2. Show that $2\mathbb{Z}$, with ordinary integer addition and multiplication, is a ring without unity.

Solution. Even integers are closed under addition and additive inverses, contain 0, and inherit the abelian group laws of $\mathbb{Z}$. If $a = 2r$ and $b = 2s$, then $ab = 4rs = 2(2rs)$ is even, so multiplication is closed. Associativity, commutativity, and both distributive laws are inherited from $\mathbb{Z}$. Thus $2\mathbb{Z}$ is a commutative ring. If it had a unity $u \in 2\mathbb{Z}$, then $u \cdot 2 = 2$. This equality in $\mathbb{Z}$ forces $u = 1$, but $1 \notin 2\mathbb{Z}$. Hence the ring has no unity.

7.3 Residue rings

Exercise 7.3. For a positive integer n, give $\mathbb{Z}_n = \{[0]_n, [1]_n, \ldots, [n-1]_n\}$ the operations

$$[a]_n + [b]_n = [a+b]_n, \qquad [a]_n [b]_n = [ab]_n.$$

Prove that they are well-defined and give a commutative ring with unity.

Solution. The representative-independence calculations and the additive group laws were established when the operations on $\mathbb{Z}_n$ were defined. It remains to verify the multiplication and its compatibility with addition. Multiplication is associative since

$$([a]_n[b]_n)[c]_n = [(ab)c]_n = [a(bc)]_n = [a]_n([b]_n[c]_n).$$

The distributive laws follow from

$$[a]_n([b]_n + [c]_n) = [a(b+c)]_n = [ab+ac]_n = [a]_n[b]_n + [a]_n[c]_n,$$
$$([a]_n + [b]_n)[c]_n = [(a+b)c]_n = [ac+bc]_n = [a]_n[c]_n + [b]_n[c]_n.$$

Commutativity follows from $[ab]_n = [ba]_n$, and the unity is $[1]_n$. In representatives $0, \ldots, n-1$, both operations are performed by taking the usual sum or product and then its remainder modulo n. When $n = 1$, there is one element and $[1]_1 = [0]_1$; this is the zero ring, which is allowed by the ring definition above.

7.4 Integer multiples in a ring

Definition 7.5. For an integer n and an element a of a ring R, the notation na denotes a multiple in the additive group:

$$na = \begin{cases} \underbrace{a + \cdots + a}_{n \text{ summands}}, & n > 0, \\ 0, & n = 0, \\ \underbrace{(-a) + \cdots + (-a)}_{-n \text{ summands}}, & n < 0. \end{cases}$$

Here n is an integer indexing repeated addition; its definition does not require a unity in R. If R has a unity, $n1_R$ denotes the corresponding multiple of 1_R.

Chapter 8

Congruences and Polynomial Arithmetic

8.1 Euler's theorem and linear congruences

For a positive integer n, write $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$ and

$$\mathbb{Z}_n^\times = \{[a] \in \mathbb{Z}_n : \gcd(a, n) = 1\}, \qquad \varphi(n) = |\mathbb{Z}_n^\times|.$$

The units form a group under multiplication. Indeed, Bézout's identity shows that $[a]$ has a multiplicative inverse precisely when $\gcd(a, n) = 1$.

Exercise 8.1 (Euler's theorem)**.** Let $a \in \mathbb{Z}$ and $n > 0$ with $\gcd(a, n) = 1$. Prove that

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Deduce Fermat's theorem: if p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Solution. For $n > 1$, the order of $[a]$ in the finite group $\mathbb{Z}_n^\times$ divides the order $\varphi(n)$ of that group, by Lagrange's theorem. Hence $[a]^{\varphi(n)} = [1]$. For $n = 1$ every pair of integers is congruent modulo 1, so the assertion also holds. If p is prime, its nonzero residue classes are exactly its units, giving $\varphi(p) = p - 1$ and the second assertion.

A linear congruence is an equation

$$ax \equiv b \pmod{n}.$$

Its solutions are residue classes modulo n, although each such class contains infinitely many integer representatives.

Example 8.1. The congruence $2x \equiv 1 \pmod 3$ has the single solution class $[2]$; its integer solutions are $x = 2 + 3k$, $k \in \mathbb{Z}$. The congruence $2x \equiv 1 \pmod 4$ has no solution, since an even integer cannot differ from 1 by a multiple of 4. For $2x \equiv 0 \pmod 4$, precisely the even integers solve the equation, giving the two solution classes $[0]$ and $[2]$ in $\mathbb{Z}_4$.

Theorem 8.2. *If* $\gcd(a, n) = 1$*, then* $ax \equiv b \pmod n$ *has exactly one solution in* $\mathbb{Z}_n$*.*

Proof. The class $[a]$ is a unit. Thus $[a]^{-1}[b]$ is a solution. If $[a][x_1] = [b] = [a][x_2]$, multiplication by $[a]^{-1}$ gives $[x_1] = [x_2]$.

Theorem 8.3. *Let $a, b \in \mathbb{Z}$, $n > 0$, and $d = \gcd(a, n)$. The congruence $ax \equiv b \pmod{n}$ is solvable if and only if $d \mid b$.*

Proof. If s is a solution, then $as - b = nt$ for some $t \in \mathbb{Z}$, so $b = as - nt$ is divisible by d. Conversely, suppose $d \mid b$ and write $a = da_1$, $b = db_1$, $n = dn_1$. Since $\gcd(a_1, n_1) = 1$, there is a solution s of $a_1 x \equiv b_1 \pmod{n_1}$. Multiplying $a_1 s - b_1 = n_1 t$ by d shows $as - b = nt$, as required.

Exercise 8.2 (Counting the solution classes)**.** Under these hypotheses, prove that if $d \mid b$, there are exactly d solutions in $\mathbb{Z}_n$. Describe them explicitly from one solution s.

Solution. The original congruence is equivalent to $a_1 x \equiv b_1 \pmod{n_1}$, which has a unique solution class $[s]$ modulo n_1. Thus all integer solutions have the form $s + kn_1$. Modulo $n = dn_1$, these give precisely

$$[s],\ [s + n_1], \ldots, [s + (d-1)n_1].$$

Two entries with indices i, j agree modulo n exactly when $dn_1 \mid (i - j)n_1$, or $d \mid i - j$. They are therefore distinct for $0 \le i, j < d$, and every other index is congruent to one of these indices modulo d.

Example 8.4. For $12x \equiv 9 \pmod{18}$, the greatest common divisor is 6, which does not divide 9. There is no solution.

Exercise 8.3. Solve $15x \equiv 9 \pmod{18}$ completely.

Solution. Here $d = 3$. Dividing the congruence by 3 gives $5x \equiv 3 \pmod{6}$. Since $5^{-1} \equiv 5 \pmod{6}$, this becomes $x \equiv 15 \equiv 3 \pmod{6}$. The three solution classes modulo 18 are

$$[3], \quad [9], \quad [15].$$

Indeed, their products with 15 are 45, 135, and 225, each congruent to 9 modulo 18.

8.2 Coefficient rings and formal polynomials

Throughout the polynomial and ideal sections, rings are commutative with identity $1 \neq 0$. Recall that the additive operation makes a ring an abelian group, multiplication is associative with identity, and multiplication distributes over addition. An *integral domain* is such a ring in which $ab = 0$ implies $a = 0$ or $b = 0$. A *field* is such a ring in which every nonzero element is a unit. Every field is an integral domain: if $a \neq 0$ and $ab = 0$, multiplication by a^{-1} gives $b = 0$. In particular, $\mathbb{Z}_p$ is a field when p is prime, by Bézout's identity. We also write $\mathbb{F}_p$ for this field.

Definition 8.5. A polynomial over a ring R is a formal expression

$$f(x) = a_0 + a_1 x + \cdots + a_n x^n, \qquad a_i \in R.$$

Two polynomials are equal when all their corresponding coefficients are equal, after adjoining zero coefficients if necessary. The set of polynomials over R is denoted by $R[x]$. For $f \neq 0$, the largest index n for which $a_n \neq 0$ is its *degree*, written $\deg f = n$; a_n is its leading coefficient. The polynomial is *monic* if $a_n = 1$. The zero polynomial has no degree in this convention.

Addition is coefficientwise, and multiplication is defined by convolution:

$$\left(\sum_i a_i x^i\right) + \left(\sum_i b_i x^i\right) = \sum_i (a_i + b_i) x^i, \qquad \left(\sum_i a_i x^i\right)\left(\sum_j b_j x^j\right) = \sum_k \left(\sum_{i+j=k} a_i b_j\right) x^k.$$

All sums here have only finitely many nonzero terms.

Exercise 8.4. Prove that these operations make $R[x]$ a commutative ring with identity. If R is an integral domain, prove that for nonzero $f, g \in R[x]$,

$$\deg(fg) = \deg f + \deg g,$$

and deduce that $R[x]$ is an integral domain.

Solution. Coefficientwise addition inherits associativity, commutativity, the zero element, and additive inverses from R. If f, g, h have respective coefficients a_i, b_j, c_ℓ, the coefficient of x^k in either $(fg)h$ or $f(gh)$ is $\sum_{i+j+\ell=k} a_i b_j c_\ell$, so multiplication is associative. Commutativity follows by interchanging i and j, and distributivity follows by distributing inside each coefficient sum. The constant polynomial 1 is a multiplicative identity.

If $\deg f = m$ and $\deg g = n$, with leading coefficients a_m, b_n, then the coefficient of x^{m+n} in fg is $a_m b_n$. In a domain this is nonzero, and no higher coefficient can be nonzero. Hence $\deg(fg) = m+n$ and $fg \neq 0$. Thus $R[x]$ has no zero divisors.

Definition 8.6. If F is a subfield of a field E, a polynomial $f(x) = \sum_i a_i x^i \in F[x]$ can be evaluated at $\alpha \in E$ by $f(\alpha) = \sum_i a_i \alpha^i$. The element α is a *root*, or *zero*, of f if $f(\alpha) = 0$. For fixed α, evaluation preserves sums, products, and the identity. In particular, $f(1) = \sum_i a_i$.

Example 8.7. The polynomial $x^2 + 1 \in \mathbb{R}[x]$ has no root in $\mathbb{R}$, but has roots i and $-i$ in $\mathbb{C}$. In $\mathbb{F}_5$, the four elements $1, 2, 3, 4$ are roots of $x^4 - 1$: their fourth powers are respectively $1, 16, 81, 256$, all equal to 1 modulo 5.

Definition 8.8. For $f, g \in F[x]$ with $g \neq 0$, say that g *divides* f, written $g \mid f$, if $f = gh$ for some $h \in F[x]$.

Example 8.9. The identity

$$x^2 - 5x + 6 = (x - 2)(x - 3)$$

shows that $x - 2$ divides $x^2 - 5x + 6$ in $\mathbb{Q}[x]$ and in $\mathbb{R}[x]$.

8.3 Division, remainders, and roots

Theorem 8.10 (Polynomial division)**.** *Let F be a field and $f, g \in F[x]$ with $g \neq 0$. There are unique $q, r \in F[x]$ such that*

$$f = qg + r, \qquad r = 0 \text{ or } \deg r < \deg g.$$

Proof. If $f = 0$, take $q = r = 0$. Write $\deg g = n$, with leading coefficient $b_n \neq 0$. If $f \neq 0$ and $\deg f < n$, take $q = 0$, $r = f$. For the remaining case use strong induction on $m = \deg f$. If $m \geq n$ and the leading coefficient of f is a_m, set

$$f_1 = f - \frac{a_m}{b_n} x^{m-n} g.$$

Its leading term cancels, so $f_1 = 0$ or $\deg f_1 < m$. The induction hypothesis, or the zero case, gives $f_1 = q_1 g + r$ with the required bound on r. Hence

$$f = \left(\frac{a_m}{b_n} x^{m-n} + q_1\right) g + r.$$

For uniqueness, suppose $f = qg + r = q'g + r'$ with both remainder bounds. Then $(q - q')g = r' - r$. If $q - q' \neq 0$, the left side has degree at least $\deg g$, whereas the right side is zero or has degree less than $\deg g$, a contradiction. Thus $q = q'$ and then $r = r'$.

Exercise 8.5. Divide $x^3 - 1$ by $x^2 - 2x + 1$ over $\mathbb{Q}$.

Solution. Subtracting $x(x^2 - 2x + 1)$ from $x^3 - 1$ leaves $2x^2 - x - 1$. Subtracting $2(x^2 - 2x + 1)$ from this leaves $3x - 3$. Therefore

$$x^3 - 1 = (x + 2)(x^2 - 2x + 1) + (3x - 3).$$

The quotient is $x + 2$ and the remainder is $3x - 3$, whose degree is less than 2.

Exercise 8.6 (Remainder and factor theorems)**.** For $f \in F[x]$ and $a \in F$, prove that

$$f(x) = q(x)(x - a) + f(a)$$

for some $q \in F[x]$. Deduce that a is a root of f if and only if $x - a$ divides f.

Solution. Division by $x - a$ gives $f = q(x)(x - a) + r$, where r is constant (possibly zero). Evaluating at a gives $r = f(a)$. Thus $f(a) = 0$ implies $f = q(x)(x - a)$. Conversely, any such factorization gives $f(a) = q(a)(a - a) = 0$.

Exercise 8.7 (Number of roots)**.** Prove that a nonzero polynomial of degree n over a field has at most n distinct roots in any field containing its coefficients.

Solution. Work in the larger field and induct on n. A nonzero constant has no roots. If a polynomial of positive degree has no root, the bound holds. Otherwise choose a root a and factor $f = (x - a)g$, where $\deg g = n - 1$. If $b \neq a$ is another root, then $0 = f(b) = (b - a)g(b)$. Since $b - a \neq 0$ in a field, $g(b) = 0$. By induction there are at most $n - 1$ such other roots, giving at most n roots altogether.

8.4 Finite multiplicative subgroups of fields

Theorem 8.11. *Every finite subgroup G of the multiplicative group $F^\times$ of a field F is cyclic. In particular, the multiplicative group of a finite field is cyclic.*

Proof. The group G is abelian. By the structure theorem for finite abelian groups, it is isomorphic to a product

$$\mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_k},$$

where each $d_i > 1$ is a prime power; the trivial group is already cyclic. Put $m = \operatorname{lcm}(d_1, \ldots, d_k)$. Every element of the product has order dividing m, and the tuple whose coordinates generate all the cyclic factors has order exactly m. Thus m is the maximum element order in G, and every $g \in G$ satisfies $g^m = 1$.

All elements of G are therefore roots of the nonzero polynomial $x^m - 1$. The root bound gives

$$d_1 \cdots d_k = |G| \le m \le d_1 \cdots d_k.$$

Consequently $m = |G|$. The element of order m generates G. Equivalently, equality of the least common multiple and the product forces the d_i to be pairwise relatively prime, and their cyclic product is cyclic. Applying the result to $G = F^\times$ proves the finite-field assertion.

8.5 Irreducibility and the coefficient field

Definition 8.12. A nonconstant polynomial $f \in F[x]$ is *irreducible over* F if it cannot be written as a product of two polynomials of positive degree in $F[x]$. Otherwise it is *reducible over* F. Thus the factors in a reducibility test must both have degree strictly less than $\deg f$. Nonzero constant polynomials are units, and are not called irreducible.

Example 8.13. Factorization depends on the coefficient ring. The polynomial $x^2 + 1$ has no factorization into positive-degree polynomials in $\mathbb{R}[x]$: such a factorization would contain a linear factor, hence a real root, whereas $a^2 + 1 > 0$ for every real a. Consequently it has no such factorization in $\mathbb{Q}[x]$ or $\mathbb{Z}[x]$ either. Over $\mathbb{C}$ it factors as $(x - i)(x + i)$. Over $\mathbb{F}_2$, by contrast,

$$x^2 + 1 = (x + 1)^2,$$

since $2x = 0$.

Exercise 8.8. Does $x + 1$ divide $x^2 + 1$ in $\mathbb{F}_2[x]$?

Solution. Yes. Direct multiplication gives $(x + 1)(x + 1) = x^2 + 2x + 1 = x^2 + 1$, so the quotient is $x + 1$ and the remainder is zero. Equivalently, evaluation at $1 = -1$ gives zero.

Exercise 8.9 (Polynomials of degree two or three)**.** Let $f \in F[x]$ have degree 2 or 3. Prove that f is reducible over F if and only if it has a root in F.

Solution. A root $a \in F$ gives a factorization $f = (x - a)q$ with $\deg q = \deg f - 1 \ge 1$. Conversely, if $f = gh$ with both degrees positive, their degrees sum to 2 or 3, so one factor is linear. Writing that factor as $cx + d$ with $c \neq 0$, its root $-d/c$ is a root of f.

Exercise 8.10 (Gauss's lemma and rational factorization)**.** Call a nonzero polynomial in $\mathbb{Z}[x]$ *primitive* if the greatest common divisor of its coefficients is 1. Prove that the product of two primitive polynomials is primitive. Deduce that a nonzero integer polynomial has a factorization in $\mathbb{Q}[x]$ into factors of positive degrees r, s if and only if it has a factorization in $\mathbb{Z}[x]$ into factors of the same degrees.

Solution. Suppose $g, h \in \mathbb{Z}[x]$ are primitive but gh is not. A prime p then divides every coefficient of gh. Reducing modulo p gives $\overline{g}\,\overline{h} = 0$ in $\mathbb{F}_p[x]$. Since this is an integral domain, $\overline{g} = 0$ or $\overline{h} = 0$. Then p divides every coefficient of one factor, contradicting primitivity.

For a nonzero integer polynomial f, let $c(f) > 0$ be the greatest common divisor of its coefficients, so $f = c(f) f_0$ with f_0 primitive. Suppose $f = uv$ in $\mathbb{Q}[x]$. Clearing denominators in each factor and removing the greatest common divisor of its integer coefficients expresses $u = \alpha g$, $v = \beta h$, where $\alpha, \beta \in \mathbb{Q}^\times$ and $g, h \in \mathbb{Z}[x]$ are primitive of degrees r, s. Thus $f = tgh$ with $t = \alpha\beta$. Write

$t = A/B$ in lowest terms, $B > 0$. Since all coefficients of $(A/B)gh$ are integers, B divides every coefficient of gh. Primitivity of gh forces $B = 1$. Therefore $t \in \mathbb{Z}$, and $f = (tg)h$ is the desired integer factorization. The reverse direction uses the inclusion $\mathbb{Z}[x] \subseteq \mathbb{Q}[x]$.

Exercise 8.11 (Rational roots of a monic integer polynomial)**.** Let

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{Z}[x], \qquad n \geq 1.$$

Prove that every rational root is an integer, and that every integer root m divides a_0 when $m \neq 0$. Determine when 0 is a root.

Solution. Write a rational root as u/v in lowest terms, with $v > 0$. Multiplication of $f(u/v) = 0$ by v^n gives

$$u^n + a_{n-1}u^{n-1}v + \cdots + a_1 uv^{n-1} + a_0 v^n = 0.$$

Hence $v \mid u^n$. Since $\gcd(u, v) = 1$, this forces $v = 1$, so the root is an integer m. Now

$$a_0 = -m\big(m^{n-1} + a_{n-1}m^{n-2} + \cdots + a_1\big),$$

which proves the divisibility for $m \neq 0$. Finally, 0 is a root exactly when $a_0 = 0$.

Exercise 8.12. Establish irreducibility over $\mathbb{Q}$ of $x^3 + 3x + 2$ and $x^3 - 2x - 2$.

Solution. Both are monic cubics with nonzero constant term of absolute value 2. Any rational root must therefore lie among $1, -1, 2, -2$. For $f(x) = x^3 + 3x + 2$ the corresponding values are

$$f(1) = 6, \quad f(-1) = -2, \quad f(2) = 16, \quad f(-2) = -12.$$

For $g(x) = x^3 - 2x - 2$ they are

$$g(1) = -3, \quad g(-1) = -1, \quad g(2) = 2, \quad g(-2) = -6.$$

Neither polynomial has a rational root, so both are irreducible over $\mathbb{Q}$ by the cubic criterion. They are primitive; Gauss's lemma also rules out factorizations in $\mathbb{Z}[x]$ into positive-degree factors. Nor can they have an integer nonunit constant factor, since they are monic. Thus both are irreducible in $\mathbb{Z}[x]$ as well.

Exercise 8.13 (Eisenstein's criterion)**.** Let $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$, with $n \geq 1$ and $a_n \neq 0$. Suppose that for some prime p,

$$p \nmid a_n, \qquad p \mid a_i \ (0 \leq i < n), \qquad p^2 \nmid a_0.$$

Prove that f is irreducible over $\mathbb{Q}$.

Solution. Suppose instead that $f = gh$ in $\mathbb{Q}[x]$, with both degrees positive. Gauss's lemma permits such a factorization in $\mathbb{Z}[x]$ with the same degrees. The product of the leading coefficients is a_n, so neither leading coefficient is divisible by p. The reductions $\overline{g}, \overline{h}$ in $\mathbb{F}_p[x]$ therefore retain their positive degrees and satisfy

$$\overline{g}\,\overline{h} = \overline{a_n}x^n.$$

Write $\overline{g} = x^r u$, $\overline{h} = x^s v$, where the constant terms of u and v are nonzero. Comparison of the least powers of x with nonzero coefficients gives $r + s = n$. Comparison of degrees then gives $\deg u + \deg v = 0$, so u, v are constants. Thus both reductions are monomials of positive degree. Their constant terms vanish, meaning that p divides the constant terms of both g and h. This implies $p^2 \mid a_0$, a contradiction.

8.6 Polynomials and polynomial functions

Note 8.14. A formal polynomial is determined by its coefficients. Its associated polynomial function is determined by its values, and these notions need not identify the same objects. In $\mathbb{F}_2[x]$ the polynomial

$$h(x) = x^2 + x$$

is nonzero, because both its quadratic and linear coefficients are 1. Nevertheless $h(0) = 0$ and $h(1) = 1 + 1 = 0$. Its associated function $\mathbb{F}_2 \to \mathbb{F}_2$ is therefore the zero function, the same function represented by the zero polynomial.

There is no conflict with the root bound: this degree-two polynomial has exactly two roots. Over an infinite field, two polynomials that define the same function must be equal, since their difference would otherwise be a nonzero polynomial with infinitely many roots.

Chapter 9

Ideals and Quotient Rings

9.1 Ideals and the quotient construction

Definition 9.1. An *ideal* I of a commutative ring R is an additive subgroup such that $ra \in I$ for every $r \in R$ and $a \in I$. It is *proper* when $I \neq R$. The quotient ring consists of the additive cosets $a + I$, with operations

$$(a + I) + (b + I) = (a + b) + I, \qquad (a + I)(b + I) = ab + I.$$

Its zero and identity are I and $1 + I$. For example, if $a - a' \in I$ and $b - b' \in I$, then $ab - a'b' = (a-a')b+a'(b-b') \in I$, so multiplication is independent of representatives. In particular, $a+I = 0+I$ precisely when $a \in I$.

Exercise 9.1. Show that an ideal containing 1 equals R. Deduce that the only ideals in a field F are (0) and F.

Solution. If $1 \in I$, then every $r \in R$ lies in I because $r = r \cdot 1$. Now let I be a nonzero ideal in a field and choose $0 \neq a \in I$. Then $a^{-1}a = 1 \in I$, so $I = F$. The zero ideal is always an ideal.

The conditions that a quotient be a field or an integral domain are expressed by maximal and prime ideals, respectively.

9.2 Maximal ideals

Definition 9.2. A proper ideal M of R is *maximal* if there is no ideal J such that $M \subsetneq J \subsetneq R$.

Theorem 9.3. *A proper ideal M is maximal if and only if R/M is a field.*

Proof. First suppose M is maximal. Since $1 \notin M$, the quotient has distinct zero and identity. Let $a + M$ be nonzero, so $a \notin M$. Consider

$$J = M + (a) = \{b + ar : b \in M,\ r \in R\}.$$

This is an ideal: it contains zero, is closed under differences, and for $s \in R$ the product $s(b + ar) = sb + a(sr)$ lies in the same set. It contains M and contains $a \notin M$, so maximality gives $J = R$. Consequently $1 = b + ar$ for some $b \in M$ and $r \in R$. In the quotient,

$$(a + M)(r + M) = 1 + M.$$

Every nonzero element has an inverse, so R/M is a field.

Exercise 9.2. Complete the converse: if R/M is a field, prove that M is maximal.

Solution. Let J be an ideal properly containing M, and choose $a \in J \setminus M$. The nonzero class $a + M$ has an inverse $r + M$, so $ar - 1 \in M \subseteq J$. Also $ar \in J$ because $a \in J$. Subtracting gives $1 \in J$, and therefore $J = R$. Thus there is no proper ideal strictly between M and R, which is maximality.

Example 9.4. For a prime integer p, the quotient $\mathbb{Z}/p\mathbb{Z}$ is a field. Hence $p\mathbb{Z}$ is maximal in $\mathbb{Z}$: an ideal J satisfying $p\mathbb{Z} \subseteq J \subseteq \mathbb{Z}$ must equal one of the two endpoints.

Exercise 9.3. Prove that R is a field if and only if it has no nonzero proper ideal.

Solution. The forward implication follows because every nonzero ideal in a field contains 1. Conversely, if there is no nonzero proper ideal, the zero ideal is maximal: every ideal properly containing it must be R. The maximal-ideal characterization implies that $R/(0)$ is a field. Since $R/(0)$ is naturally isomorphic to R, the ring R is a field.

9.3 Prime ideals

Definition 9.5. A proper ideal P of R is *prime* if, for all $a, b \in R$,

$$ab \in P \quad \Longrightarrow \quad a \in P \text{ or } b \in P.$$

Example 9.6. If p is a prime integer, then $p\mathbb{Z}$ is a prime ideal of $\mathbb{Z}$. Indeed, $ab \in p\mathbb{Z}$ says $p \mid ab$, so $p \mid a$ or $p \mid b$, equivalently $a \in p\mathbb{Z}$ or $b \in p\mathbb{Z}$.

Exercise 9.4. For a proper ideal P of R, prove that P is prime if and only if R/P is an integral domain.

Solution. The quotient is a commutative ring with identity, and $1 + P \neq 0 + P$ because P is proper. If P is prime and $(a + P)(b + P) = 0 + P$, then $ab \in P$, so $a \in P$ or $b \in P$. Thus one of the factors is zero, proving that R/P is a domain. Conversely, if R/P is a domain and $ab \in P$, then $(a + P)(b + P) = 0 + P$ implies $a + P = 0 + P$ or $b + P = 0 + P$. This is precisely the prime-ideal condition.

Exercise 9.5. Prove that every maximal ideal is prime.

Solution. If M is maximal, then R/M is a field. Every field is an integral domain, so the preceding characterization implies that M is prime.

9.4 Principal ideals in the integers and in polynomial rings

Definition 9.7. For $a \in R$, the *principal ideal generated by* a is

$$(a) = \{ra : r \in R\}.$$

It is an ideal: it is closed under differences, and multiplication by any ring element preserves the set. In particular, for $n \in \mathbb{Z}$, $(n) = n\mathbb{Z}$.

Exercise 9.6. Prove that every ideal of $\mathbb{Z}$ is principal.

Solution. The zero ideal is (0). If $I \neq (0)$, then I contains a positive integer, since it contains the negative of each of its elements. Let d be its least positive element. For $a \in I$, integer division gives $a = qd + r$ with $0 \leq r < d$. Since $r = a - qd \in I$, minimality forces $r = 0$. Thus $a \in (d)$, proving $I \subseteq (d)$. The reverse inclusion holds because $d \in I$ and I is an ideal.

Exercise 9.7. Let F be a field. Prove that every ideal of $F[x]$ is principal.

Solution. The zero ideal is generated by the zero polynomial. If $I \neq (0)$, choose a nonzero $g \in I$ of least degree. Since $g \in I$, all its polynomial multiples lie in I, so $(g) \subseteq I$. For any $f \in I$, division gives

$$f = qg + r, \qquad r = 0 \text{ or } \deg r < \deg g.$$

But $r = f - qg \in I$, so a nonzero r would contradict the minimal choice of $\deg g$. Therefore $r = 0$, showing $f \in (g)$ and $I = (g)$. Multiplying g by the inverse of its leading coefficient gives a monic generator of the same ideal if desired.

Exercise 9.8. If $p(x) \in F[x]$ is irreducible, prove that

$$(p(x)) = \{p(x)f(x) : f(x) \in F[x]\}$$

is a prime ideal.

Solution. First (p) is proper: an equality $1 = ph$ would contradict the degree formula, since $\deg p \geq 1$. In fact (p) is maximal. To see this, let J be an ideal with $(p) \subseteq J \subseteq F[x]$. By the principal-ideal result, $J = (g)$ for some $g \neq 0$, since $p \in J$. Then $p = gh$ for some $h \in F[x]$. Irreducibility of p implies that g or h is a nonzero constant. If g is constant, it is a unit and $J = F[x]$. If h is constant, it is a unit and $(g) = (p)$. Thus (p) is maximal, and hence prime. In particular, $ab \in (p)$ forces $a \in (p)$ or $b \in (p)$, as required.